

OFFSTAT MIDSEM NOTES

LEC 1

1 National Accounts - An Introduction

2 Contents of Session 1

Introduction – National Accounts and Economics – Social Accounting and National Accounts – Realm of Economics. Evolution of Conceptual Framework of Macroeconomics. Measuring Production. Circular flow.

3 Introduction

3.1 National Accounts and Economics

3.2 Social Accounting and National Accounts

Richard Stone, in his Nobel Memorial Lecture entitled “The Accounts of Society” (1984), presented how accounting can be useful in describing and understanding society. Accounting for the society or Social Accounting, in his view, ought to rest on three pillars – studies of economic, socio-demographic and environmental phenomena. In his perception, economic accounts of the society, called National Income Accounting, forms only a part of the Social Accounting. Main focus: The System of National Accounts (SNA) – its latest version 2025 SNA and the earlier version 2008 SNA.

3.3 National Accounts

J.R. Hicks: Like the study of the human body, Economics has two parts: Economic physiology – deals with the way the economic system works. i.e. economic theories; and Economic anatomy – understanding the structure of the economy, i.e. estimates of macro-economic aggregates or national accounts statistics.

3.4 Economics – a definition

“Economics is the study of how people and society end up choosing, with or without the use of money, to employ scarce productive resources that could have alternative uses – to produce various commodities and distribute them for consumption, now or in the future, among various persons and groups in society” – Samuelson (1980)

3.5 A re-look at the definition

In the definition, “choice of use” refer to human acts of Production: creation of goods & services with human effort. Consumption: use of produced goods & services for human satisfaction. Accumulation: saving goods & services for future consumption and further production. and “scarce resources” refer to only those that have value in exchange.

3.6 Value in Exchange

To exchange one thing for another is a trait unique to human beings. Value of an object can be judged by its use to human beings or by its power of purchasing other objects. The latter is referred to as value in exchange.

3.7 Economic Entities

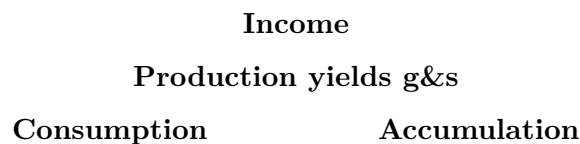
Economic entities or economic agents are natural or legal units like Households, Companies, cooperatives, trusts etc., Government, Non-profit institutions (NPIs). Only the goods & services (and money) that have value in exchange and their between-entity changes in ownership are considered in economics. Note that when a household carries out production, it is treated as two separate units – a production unit and a consumption unit.

3.8 Realm of Economics

Economics deals with only the human acts that involve between-entity exchange (and transfers) of objects (money, goods & services) that have value in exchange, and only the volume of objects with value in exchange and the events that bring about changes in that like production (by definition with human effort), consumption: final consumption, intermediate consumption (IC) and depreciation & depletion, discoveries of natural, non-produced assets like minerals, destruction of property by human-made and natural calamities. Economics also reckons all kinds of between-entity flow of money.

3.9 Central role of production

The human act of production is carried out for sustenance and creation of wealth. Production entails (productive) “work” or “labour”.



4 Evolution of Conceptual Framework of Macroeconomics

4.1 Economies before Economics (1)

Economics, as a subject of study, is of recent origin, but every human lifestyle since the prehistoric times had an economy. Question: Did the earliest human lifestyle of ‘hunting and gathering’ have an economy? How would you define production, consumption and accumulation for such a lifestyle? Ans: Production: Hunting & gathering and tools making. Accumulation: None? Another question: How would you describe these three human acts for pastoral life? Ans: Production: Domestication of animals and animal husbandry. Accumulation: Domesticated animals.

4.2 Economies before Economics (2)

About 12 thousand years ago, pastoral lifestyle, characterised by domestication and rearing of animals, gave way to agrarian lifestyle when they learnt to domesticate plants. Cultivated natural resources became economic assets and people adopted the settled way of life. From the agrarian societies emerged civilizations with different forms of feudalism. Peoples’ conception of social lives was constrained by authority (King and the feudal lords) and faith (Church). Theologians, such as St. Thomas Aquinas (13th century), held trading and usury immoral or unethical activity. Feudalism continued for centuries, till the 13th century.

4.3 Rise of Mercantile groups in Europe

During 13th to 16th century, Europe saw the beginnings of industrial innovations and growth of trade and commerce following the early maritime explorations. The nation states came into existence, mercantile groups emerged and feudalism declined. The merchant class played a significant role when Europe was undergoing commercial revolution, which soon gave way to industrial revolution. Merchants helped in acquiring the primitive accumulation. But unrestrained power in the hands of autocratic rulers and domination of theocracy in social life persisted.

4.4 Mercantilism (17th and 18th century)

By the 17th century, the mercantile groups acquired considerable power. As explorers and armies conquered new lands, rulers gave merchants the right to trade with the territories. The alliance between the rulers and the merchants came to be known as Mercantilism or bullionism – the earliest prescription for prosperity of nation states – laid the foundations of economics as a field of study. Most of the Mercantilist writers were merchants or those representing interests of merchants – administrators or ministers. They turned away from medieval theocracy and moved towards reason and science.

4.5 Mercantilist doctrine

Notable mercantilists: English merchant Thomas Mun, and writers William Petty, and John Locke; and French ministers Jean-Baptiste Colbert and Jacques Necker. They advocated policies, but did not have any well-developed economic theory to offer. They said that wealth is gold and silver and the state should act to protect and increase their wealth. Puzzled by the growing prosperity of the (a century and quarter old) Dutch economy, the mercantilists were drawn to uphold ‘favourable balance of trade’ as a means of achieving prosperity:

favourable balance of trade → more money supply & income → more expenditure → more demand →
more domestic production → growth and prosperity

4.6 Criticism of Mercantilists’ view

Mercantilists were obsessed with gold instead of the goods that we need to live. That money supply can increase a nation’s welfare is illusory. Accumulation of money only created inequality in the ownership of landed property and thus the wealth. Physiocrats thought trading to be a non-productive activity incapable of generating surplus. In spite of the shortcomings in mercantilists’ doctrine, ‘balance of trade’, or more generally ‘balance of payment’ is used even today as a measure of a country’s economic performance. In national accounts, the ‘net lending / borrowing’ is considered to be the end result of all monetary transactions of an economy.

4.7 Birth of Economics

Economics, as a field of study, is a product of the developments in the age of enlightenment or renaissance in Europe. In the 200 years leading to 18th century, philosophical ideas of ‘freedom’ and ‘natural order’ began to evolve, leading to a theory of ‘free market’ and laissez-faire theory. And ultimately to the theory of market. The three 18th century developments of this tendency were the laissez-faire theories of the Physiocrats in France, Giliani in Italy and Adam Smith in Scotland. They advocated ‘free market’ and minimal intervention of external agent (government). Established the core of macroeconomics.

4.8 Physiocracy

In the middle of 1750’s, Physiocrats, led by François Quesnay, really dominated the intellectual world across Europe for 15 years. 1759: François Quesnay. *Tableau économique* – laid the theoretical roots of this concept is the “theory of the economic circle”. The physiocratic version of the economy was probably the first macro-economic articulation ever in economic theory. *Tableau économique* is the precursor of Input-output transaction tables (IOTT) – a product of national accounts compilation and, more importantly, uses the concept of ‘value added’, the method of unduplicated accounting of production in an economy. But, they considered agriculture to be the only productive sector.

4.9 Adam Smith

An Inquiry into the Nature and Causes of the Wealth of Nations (1776)

Labour is the source of new wealth. Production process requires active participation of human labour – the root of economic value. But, he distinguishes between Productive labour: provides goods and generate surplus. Unproductive labour: provides non-factor services, especially personal services, and also civil services and army. Value according to cost of production = labour cost + ‘normal’ profit + rent and exchange value results from the interaction of demand and supply.

4.10 Adam Smith (contd.)

Invisible hand: the Doctrine of Market Economy

Humans are not rational egocentric machines, but are as complex as individuals of today. They are basically competitive, cooperative as well as collaborative. His ‘state’ is endogenous and not exogenous as

suggested by the earlier philosophers, such as Lock, Hobbes and Jean-Jacques Rousseau. He uses a notion of ‘impartial observer’ in every individual. This leads to advocate for ‘regulated freedom’ – *Liberté*, with rules and norms and governance; and free competitive markets and the ‘invisible hand’ that represents it so long the impartial observer in the individuals is working well.

4.11 Jean Baptiste Say

Traité d'économie politique [Treatise on Political Economics] (1817)

An ardent student, admirer and follower of Smith. Unlike Smith, Say explicitly defines the supply and demand sides. He argued that the aggregate value of all g&s produced was the same as the aggregate incomes distributed to the factors of production. The very process of production generates purchasing power. What follows from these arguments is the accounting identity of aggregate demand equals aggregate supply, in other words, expenditure is equal to income. Provided the basis of circular flow in an exchange economy.

5 Measuring Production

5.1 Measure of Production – Value Added

Alfred Marshall in his *Principles of Economics* (1890) discussed the idea of GDP, which he referred to as “Social Income”.

He states: “... We must however not count the same thing twice. If we have counted a carpet at its full value, we have already counted the values of the yarn and the labour that were used in making it; and these must not be counted again”.

5.2 Measure of Production – Value Added

For a production unit, measuring value of “production” requires netting out of the value of goods and services consumed in the process of production from the value of output. This is called value added (VA)

5.3 Gross Value Added and Gross Domestic Product (GDP)

The value of goods and services consumed in the process of production is called intermediate consumption (IC)

In other words,

$$(\text{gross}) \text{ value added (GVA)} = (\text{gross}) \text{ value of output (GVO)} - \text{IC}.$$

The sum of (gross) value added of all the production units of an economy is called the Gross Domestic Product (GDP) – represents the value of production in the economy

6 Circular Flows

6.1 Basic Circular Flow in an Exchange Economy

Two-sector Model (Real flow and money flow)

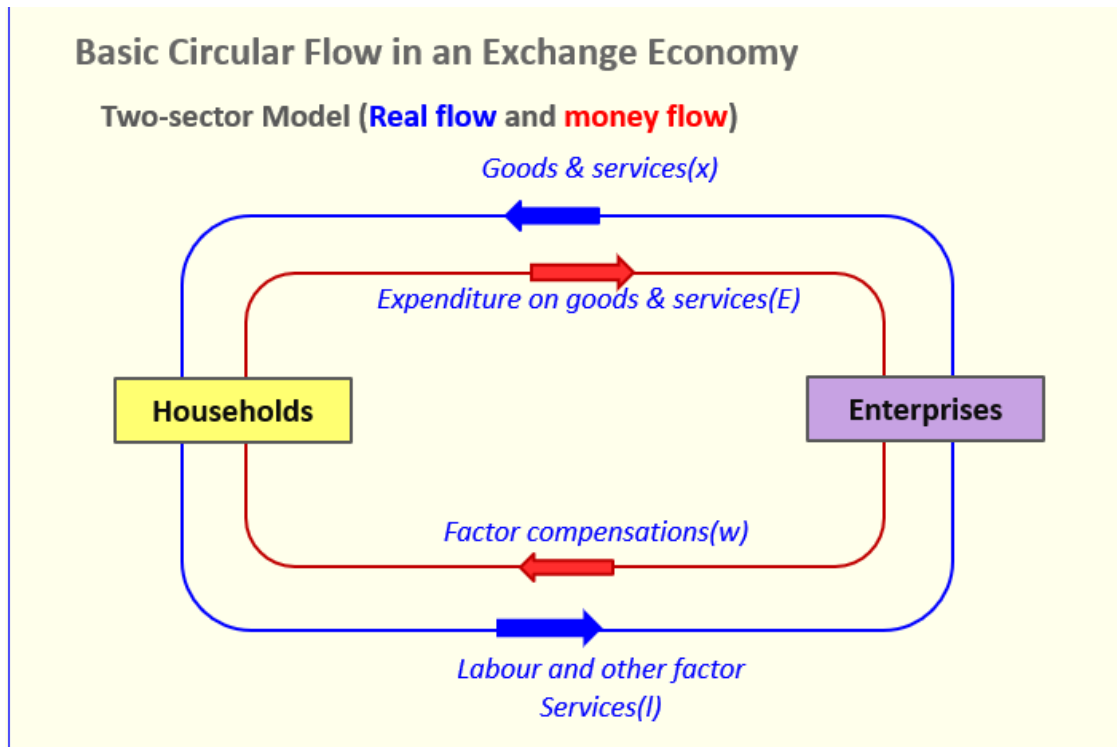


Figure 1: Circular Flow

6.2 Basic Circular flow (Contd.)

The interactions involve flows going in two directions: Flows of goods and services, called real flows. Flows of payments for these goods and services, called money flows. Households are defined as the owners of all factors of production – land, labour, capital and entrepreneurship. Households provide factor services and in return gets paid by the enterprises – wages & salaries (w & s) and operating surplus (OS). This constitutes the income of the economy.

6.3 Macro-economic Framework

The macro-economic framework is based on the premises that all goods and services produced in the domestic economy are put to “use” of one kind or the other; circular flows income and expenditure of the residents and non-residents participating in transactions of the domestic economy. The framework establishes the equivalence of supply and use of goods & services produced; the value of production of goods and services, income generated in production and expenditure on products and non-produced assets. This is the basis for ‘three approaches of measuring GDP’.

6.4 Three Approaches of Measuring GDP

Production approach: sum of GVA of all the production units in the economy

Income approach: sum of income of all the households in the economy

Expenditure approach: sum of all expenditure made on goods & services produced in the economy

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LEC 2

7 Macroeconomic Framework of SNA

8 Contents of Session 2

Evolution of Conceptual Framework of Macroeconomics (Contd.). Circular flow – a modified view. Macroeconomic Framework of SNA.

9 Evolution of Conceptual Framework of Macroeconomics (Contd.)

9.1 John Maynard Keynes – the paradigm shift

J.B.Say provided the basis of circular flow in an exchange economy and Leon Walrass, in his *Éléments d'Économie Politique Pure, ou Théorie de la richesse sociale* (1874), created the mathematical foundations to the circular flow.

They upheld *lessaize faire* or the market mechanism as the key to prosperity. This view persisted even when Alfred Marshall published his *Principles of Economics* (1890).

But, this view was based on two vital assumptions: (i) no time lag between production and income and (ii) no unspent factor income during a period of time.

Keynes brought about a paradigm shift of focus. He established that market mechanism was not efficient all the time and advocated active intervention of the government.

9.2 Birth of Macroeconomics

Macroeconomics, as we know today, was born in 1936, when Keynes published his *The General Theory of Employment, Interest, and Money*.

Keynes framed the Say's law: "supply creates its own demand" and argued that supply does not create demand. He claimed: People's spending power is determined by what they earn. Rise in effective demand leads to rise in production and supply. Prices are determined by cost of production (supply-side). Prices and wages are rigid.

He assigned a pivotal role to aggregate expenditure, which is consumption + investment.

10 Circular flow – a modified view

10.1 Use of Produced Goods and Services

The circular flow discussed earlier is oversimplified. It ignores two very basic aspects of the production process.

An enterprise uses output (goods and services) of other enterprises or of itself in its production process: Intermediate Products: raw materials, semi-processed materials and services; or Machinery, equipment, building & construction and capital services (financed by investments made by households).

This leads to a modified representation of circular flow.

10.2 National Economy Circuit

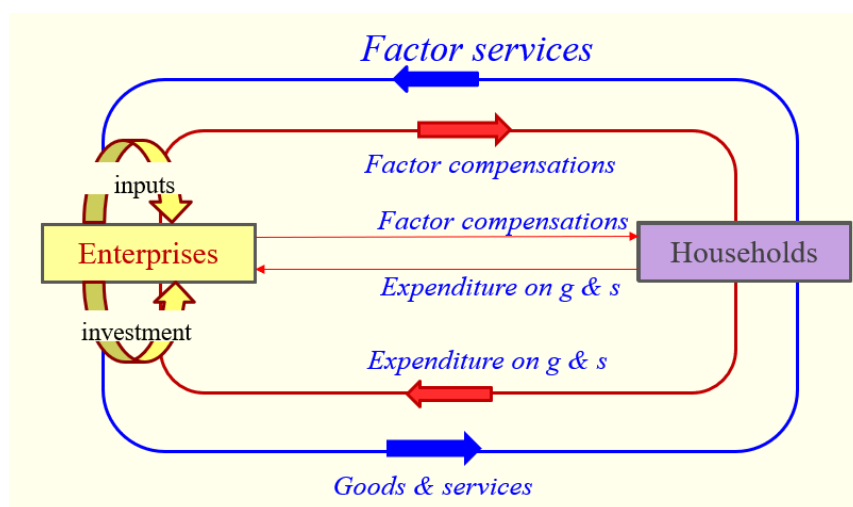


Figure 2: NEC

There are however enterprise-to-enterprise flows of g&s

Three approaches of measuring GDP: Production $\equiv$ Income $\equiv$ Expenditure

10.3 Another Illustration

In an economy with a number of enterprises, the receipts and payments of the enterprises from sale of products for household consumption & capital goods were:

Value of goods and services sold: 18,000 Rs

of which

value of inputs: 3,000 Rs

products for household consumption: 14,000 Rs &

capital goods (say tractors): 1,000 Rs

Rent paid: 1,200 Rs

Interest paid: 900 Rs

Wages and salaries paid: 8,900 Rs

Capital goods produced by one were purchased by other enterprises.

10.4 Illustration (Contd.)

With the knowledge of this detail,

What was the value of production?

What was the total income of its residents?

What was households' consumption expenditure?

What was enterprises investment expenditure on capital goods?

What was the total expenditure on purchase of goods & services of its residents?

10.5 Illustration (Contd.)

The way "production" of an economy is defined, the intermediate consumption gets eliminated and what remains are: Goods & services for final consumption and Capital goods & services for investment.

In our illustration, production = 15,000

as household consumption = 14,000

(gross) Investment = 1,000

This demonstrates the basic macro-economic relationship:

$$\text{value of production (Y)} \equiv \text{Consumption (C)} + \text{Investment (I}_g\text{)}$$

$\equiv$ final aggregate expenditure.

10.6 A consequent question

What if all the goods and services that are produced are not sold in the market?

Will the value of production still be equal to expenditure?

Note that the unsold goods goes into inventory.

The change in inventories (CII) of finished products is included in the value of production and since, the unsold goods are treated as economic assets, the CII is considered a part of investment.

Thus, the basic relations should still hold good.

10.7 Illustration with a modification

Consider the following change:

As before,

the value of goods and services produced: 18,000 Rs

of which,

value of inputs 3,000 Rs

consumption products: 14,000 Rs

and capital goods (say tractor): 1,000 Rs

But the households now spend only 13,500 Rs and put 500 Rs in their piggy banks.

10.8 Illustration with a modification (contd.)

Households spent only 13,500 Rs for consumption.

Thus, savings (S) of the households, would be 1,500 Rs

(gross) investment will now be

1,000 Rs for purchase of capital goods

plus 500 Rs for the CII.

Thus, both the identities would still hold.

$$\text{Production} \equiv \text{Income} \equiv \text{Expenditure}$$

$$(\text{gross}) \text{ Investment } (I_g) \equiv \text{Savings } (S)$$

11 Macroeconomic Framework of the SNA

11.1 National Economy Circuit

Wages & salaries flow directly to the households. But, the other factor compensations, such as interest & profits are often routed through financial institutions.

Now, let's take out the "Financial Institutions" from "Enterprises" and also introduce "Government" to see their roles in a more realistic setup.

Expenditure on g & s

Factor compensations

inputs

investment

Gross Value of Output (GVO) = Households' final consumption expenditure (HFCE) + value of inputs + value of investments

Or $GVO = HFCE + IC + \text{Gross Capital Formation (GCF)}$

Or $GVA = GVO - IC = HFCE + GCF$

which corresponds to the macro-economic identity: $Y = C + I$

11.2 National Economy Circuit

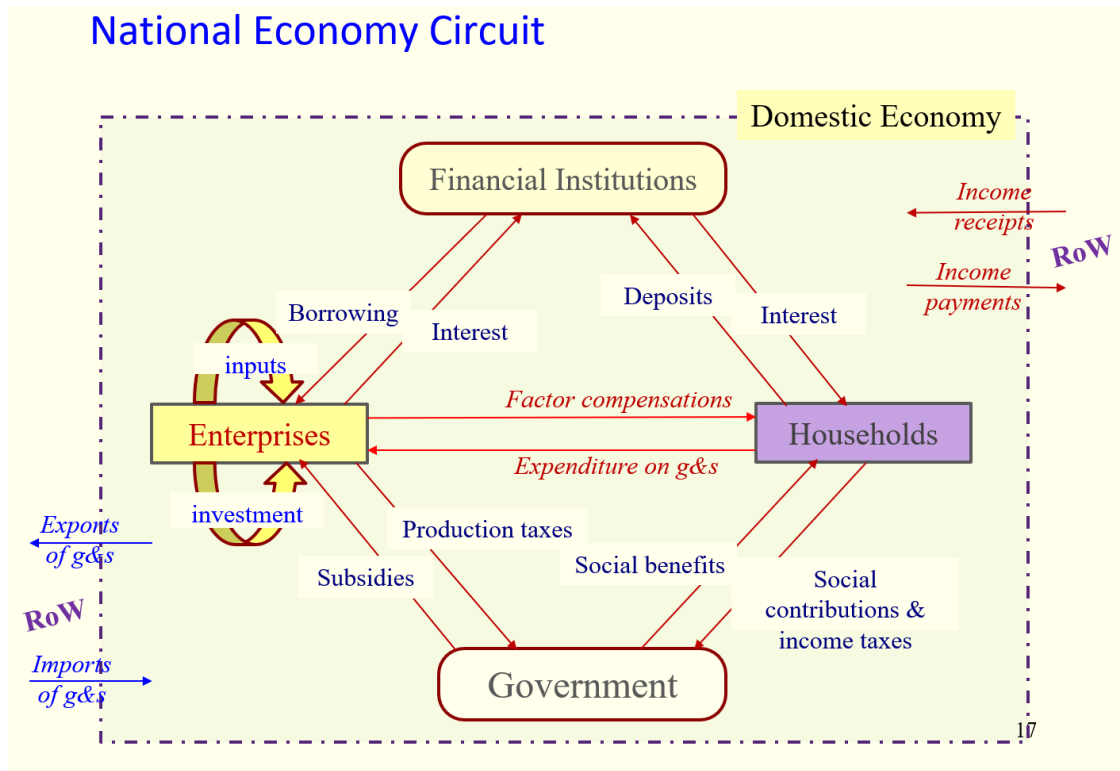


Figure 3: Caption

11.3 GVA at basic price

An enterprise's earnings from production is the GVA at basic prices

= Receipts from sale of its products

minus (all product taxes – all product subsidies)

[$\approx$ indirect taxes minus subsidies]

minus payments made for purchase of inputs (IC)

= Gross value of output at basic prices (GVO_{bp}) minus IC

GVA at basic prices, $GVA_{bp} = GVO_{bp} - IC$

11.4 Macro-economic Framework

One of the basic premises of macro-economic framework is all goods and services produced in the domestic economy are put to “use” of one kind or other. In other words, it establishes the equivalence of supply and use of goods & services.

$$\text{Domestic output (GVO)} + \text{Imports (M)} = \text{HFCE} + \text{IC} + \text{GCF} + \text{Exports (X)}$$

Household Final Consumption+Intermediate consumption

Gross Capital Formation+Exports

At the economy level, GDP is defined as (1993 / 2008 SNA), tax and subsidies

$$GDP = \sum GVA_{bp} + \text{product (t-s)} + \text{(t-s) on imports}$$

where $GVA_{bp} = GVO_{bp} - IC$

11.5 Some Questions

Which of the following are true

- (a) GDP can be measured by three approaches – production, income and expenditure.
- (b) By production approach, $GDP = GVA$ of all the production units in the economy
- (c) By income approach, $GDP =$ sum of income of all the households in the economy
- (d) By expenditure approach, $GDP =$ sum of all expenditure made on goods & services produced in the economy.
- (e) Total expenditure = final consumption expenditure + Gross Capital Formation (GCF) + exports net imports

Ans.: only (a), (b), (c) & (e).

11.6 Macro-economic Framework (1)

Applying the principles of valuation, both the sides are valued at producers' prices,

$$\Sigma GVA_{bp} + \text{product (t-s)} + \text{Imports (M)} + \text{imports duties} = HFCE + GCF + \text{Exports (X)}$$

Or

$$GDP + M + \text{import duties} = HFCE + GCF + X$$

This is called “Commodity-flow identity” in the SNA

This corresponds to the macro-economic identity

$$Y + M \equiv C + I + X$$

11.7 Macro-economic Framework (2)

Turning to the macro-economic premises of equivalence of production and income,

$$GDP \equiv \text{factor compensations from domestic production} + \text{production (t-s)} + \text{(t-s) on imports}$$

But, there are inflow from & outflow to Rest of the World (RoW) of factor compensations.

Thus, the residents' income is gross national income (GNI),

$$GNI \equiv \text{factor compensations from domestic enterprises} + \text{production (t-s)} + \text{(t-s) on imports} \\ + \text{net inflow of factor compensations from the RoW.}$$

GNI is also referred to as GNP in macro-economics.

11.8 Some More Questions

Which of the following are true

- (a) GNI includes income tax from the rest of the world (RoW) received by the government
- (b) GNI is the measure of production of an economy
- (c) GNI is equal to GDP plus (net) primary income earned (by the residents) from RoW
- (d) GNI includes money received by resident households from a non-resident relative
- (e) GNI includes receipts from consultancy services provided to the rest of the world (RoW)
- (f) GNI is not affected by the level of production in the domestic economy.

Ans.: Only (c) & (e).

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LEC 3

12 Economic Flows – macro-economic flow variables

13 Economic Flows

All changes in the volume or value of economic assets of an economic unit that are brought about by human acts, changes in price structure and natural events are defined as economic flows in the SNA.

Examples: sale & purchase of goods & services; production of goods and services; payment of taxes; payments & receipts of transfers; destruction of economic assets by natural or human-made calamities; increase in value of an asset caused by holding it over time.

14 Which of the following are economic flows?

- a. Paddy cultivation
- b. Destruction of property by calamity.
- c. Father giving money to his son, living in the same household (an economic unit)
- d. Drawing water from river for irrigating land.
- e. Growth of fish stock in a river.
- f. Price appreciation of real estate property.

[Answer: (a), (b), (d) and (f) are economic flows. (c) is not an economic flow, since within-unit flows are not economic flows. (e) Is not an economic flow, since stock of fish in a river is not an economic asset.]

15 Economic Flows – different forms

- a) all receipts and payment relating to production, consumption and accumulation. Production includes activities of trading in goods & services and financial assets
- b) transfers – such as donations, contributions, social benefits, and income taxes.
- c) extinction or emergence of economic value
- d) change in composition or value of stock of economic assets/liabilities.

The flows in categories (a) & (b) are called ‘transactions’; those in category (c) ‘volume change’ and those in category (d) are caused by ‘changes in level and structure of prices’

16 Economic Flows - classification

Economic flows

Transactions

Exchange

Transfers

Other flows

Other changes in volume

Due to changes in level & structure of prices

17 Transactions - Definition

A transaction is an economic flow that involves interaction between institutional units by mutual agreement or an action within an institutional unit that is analytically useful to treat like a transaction, often because the unit is operating in two different capacities

18 Kinds of transactions

Transactions are of two kinds:

Exchange: goods & services are exchanged between institutional units or within institutional units operating in different capacities.

Transfer: one institutional unit provides goods or service or money to another unit without receiving anything in return as counterpart.

Transactions can be actual observable flows or they can be notional flows. The notional flows are not observable and are thus estimated or imputed for accounting purposes.

19 Types of transactions

Transactions include all flows connected with

— Production and exchange of goods and services: Two kinds transactions in

Produced goods and services or products

Non-produced assets

— Distribution of income generated in production of goods & services

— Transfers: Two kinds

Capital transfers [discussed in a latter session]

Current transfers [discussed in a latter session]

— Transactions in financial assets and liabilities:

net acquisition of assets and net incurrence of liabilities.

20 Which of the following are considered as transactions? If yes, whether transfer or exchange?

a) Production of crops

b) Payment of income tax

c) Sale of smuggled goods

d) Picking co-passengers pocket

e) Payment made after losing a bet

[Answer: (a) is exchange; (b) is transfer; (c) is exchange (an illegal ; trading activity); (d) is transfer; (e) is transfer;

21 Transactions of current and capital nature

Current transactions: All transactions relating to production and final consumption and current transfers

—

production-related transactions

receipts and payments of primary income – income originating from production process

final consumption expenditure

current transfers [discussed in a latter session]

Capital transactions: All transactions relating to creation of or change in the volume of assets of an unit. These include

capital formation (investment),

acquisition and disposal of non-produced natural assets

capital taxes and

capital transfers [discussed in a latter session]

22 Transactions from other perspectives

Between and within unit transactions: Transactions may take place between institutional units or within the same institutional unit, the latter are necessarily non-monetary.

Monetary and non-monetary transactions: Transactions involving money flow are monetary and those without money flow are non-monetary. A monetary transaction is always a two-party transaction.

The non-monetary transactions are without an equivalent monetary flow. They are measured indirectly, since not observable. Commonly imputed using prevailing market value.

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LEC 4

23 Production boundary and Taxes & Subsidies

24 Contents

Production boundary

General Production Boundary

Production Boundary of 2008 & 2025 SNA

Taxes and subsidies

25 SNA Production Boundary

26 Production redefined

Production: a physical process carried out under responsibility, control and management of an institutional unit, in which labour and assets are used to transform inputs of goods and services into outputs of other goods and services.

Output of a production process: goods and services that can be sold in markets or at least be capable of being provided by the production unit to other units, with or without charge.

27 Kinds of Products

Two main kinds of output or products result from production processes:

Goods

Services and

Data

collectively, called products. Goods are physical, produced objects characterised by existence of demand, ownership rights established and transferrable ownership.

28 Kinds of output (contd.)

Services are the result of a production activity that changes the conditions of the consuming units, or facilitates the exchange of products, (such as grocers & trader) or financial assets (such as banks).

Data are the information content produced by accessing and observing phenomena; and recording, organising and storing information elements in a digital format. These are produced using labour and capital. Capitalised when used in production for more than one year. Subject to depreciation.

29 General Production Boundary

The general production boundary includes all human activities that are productive in an economic sense. These human activities include all goods-producing activities, whether or not solely for own use and all services-producing activities that can be provided to others.

In fact, it excludes only the basic human activities such as eating, drinking, sleeping, taking exercise, etc.,

that it is impossible for one person to employ another person to perform instead.

30 Production Boundary of 2008 / 2025 SNA

includes:

Production of all individual or collective goods or services that are supplied to other units or intended to be so supplied

Own-account production of the following services by households and corporate sector units:

housing services by owner-occupiers (also called “ownership of dwellings”)

domestic and personal services produced by employing paid domestic staff (such as domestic maids and cooks)

services that produces intellectual property products for own-account capital formation.

Own-account production of all goods that are retained by their producers for their own final consumption or capital formation

31 Intellectual Property Products

Intellectual property products are results of research & development activities or innovations and artistic endeavours.

Examples:

results of research and development, mineral exploration and evaluation, computer software and databases, and entertainment, literary or artistic originals.

SNA identifies four distinct and well-defined categories of IPPs :

research and development

mineral exploration and evaluation

computer software, data and databases; and

entertainment, literary and artistic originals.

32 SNA Production Boundary

Includes:

(a) Production of all goods whether for the market or not

(b) Production of services only if any of the 3 conditions are fulfilled

sold on the market

supplied to units other than producers

produced by paid labor

(c) Production of own-account housing services

(d) Production of IPP for own-account capital formation.

Excludes:

a) Factors that affect quality of life

b) Various housework, social activities, cultural activities, and voluntary services

33 A few Questions

Which of the following fall within the production boundary of SNA?

a. Maintenance and repair of household durables

b. Production of crops that are retained by the farmer for own final consumption of its household

c. Production of housing services by owner-occupiers

- d. Transportation of household members
- e. Natural growth of cultivated forests
- f. Distribution of goods whose sale, is forbidden by law, such as narcotics.

Ans: b, c, e and f fall within the SNA production boundary.

34 Production Boundary – specific cases

Production boundary also includes:

Types of goods or services that are not paid for separately at the market at economical significant prices.

Production of government services and output of other nonprofit institutions (other non-market output)

Banking services not explicitly charged for

Production of goods (but not services) for own final consumption

35 Production boundary also includes

Imputed services of owner-occupied dwellings

Domestic and personal services produced by employing paid domestic staff

Illegal and concealed production and the underground economy are included

(Note, theft is not production)

Own account production of capital goods, including intellectual property product

36 Rent and Rentals

Both are payment for getting users rights on non-financial assets:

Rent for use of non-produced non-financial assets

Examples: rent on land and other natural resources, payment made to the government by a mobile phone company for spectrum rights etc.

Rental for use of produced non-financial assets.

Examples: rental for building, machinery, transport equipment, artistic originals, patents, etc.

37 Taxes and Subsidies

38 Functions of Government

General government also carries out:

redistribution of income, through taxes and subsidies

non-market production (of government services) and

market production. The part of the government mainly engaged in providing non-market services is called 'General Government' in the SNA.

Recall that the units of the government like

national railways, airlines, electricity generation and distribution units etc.

that are run like business enterprises [i.e., engaged in market production] are treated as quasi-corporations and are included in the corporate sector.

39 Consumption of Government Services

In the expenditure approach, the entire non-market output of the general government is considered to be consumed by the general government itself.

Thus, the Government Final Consumption Expenditure,

$GFCE = \text{Total Government output} + \text{goods \& services purchased to be provided free to households} - \text{receipts from}$

40 Redistributive Activities of Government

Governments collect taxes from production units and citizens and makes subsidies only to production units.

Taxes have an effect of increasing prices of goods and services in the market.

They reduce the financial resources in the hands of the households and enterprises, and thus affect the private consumption expenditure and capital formation.

Subsidies have the opposite effects.

41 Taxes & Subsidies

Taxes are compulsory, unrequited payments made to government units by households

enterprises and

other institutional units.

Subsidies are current unrequited payments

by the government to enterprises only

for production activities

on the basis of level of production - quantities or values of the goods or services produced/ sold/ imported.

OFFSTAT MIDSEM NOTES

LEC 5

42 Domestic Economy and Production Process

42.1 Domestic Economy

42.2 What constitutes an Economy?

All institutional units residing in the economic territory of a country during the accounting period constitute its economy.

The definition involves three concepts:

- Economic territory
- Institutional Units and
- Residence.

42.3 Economic Territory

The geographic territory administered by the government of the country within which persons, goods, and capital can circulate freely. This includes:

- territories for which the country has exclusive rights
 - like airspace, territorial waters, and continental shelf in international waters,
- territorial enclaves located in other countries with political agreement with the host country.
 - like embassies, consulates, military bases, scientific stations, and information or immigration office
- Free zones, offshore activities

42.4 Institutional Unit

Definition: an economic entity that is capable of

- owning assets,
- incurring liabilities,
- carrying out economic activities taking decisions on all aspects of economic life and
- engaging in transactions with other entities.

A foundational premise of the SNA framework:

All transactions take place only through institutional units.

Institutional units are, thus, the basic units, but not the smallest units, for compilation of national accounts.

42.5 Institutional Sectors

Economy

Corporate

Financial corporate sector

Non-financial corporate sector

General government

Household sector

NPISHs

42.6 General Government

Government is made up of government units. These provide

- general administrative services
 - activities of organs of state [i.e. ministries and departments]
- provide specific services to the general population
 - external affairs and
 - defense services
 - fiscal services [through or together with the Central Bank]
 - police, jail, law and justice etc.
- provide individual services
 - health services [State-run health systems/ government hospitals]
 - education services [State-run educational institutions – schools, universities]

42.7 Economically Significant Price

- Economically significant prices are prices that have a significant effect on the amounts that producers are willing to supply and on the amounts purchasers wish to buy.
- The implication of these in practice is that the sales normally cover the majority of the production costs.

42.8 Household Sector

Households are composed of persons or group of persons with common kitchen (food arrangement).

Household sector – includes unincorporated enterprises – supplies labour – is the major final consumer – includes household operated activities – includes Informal sector

42.9 Residence

- Definition: The economic territory in which the institutional unit its centre of predominant economic interest. [2008 SNA]
- The definition of Domestic Economy is based on the concept of RESIDENCE not Currency or Nationality.
- All resident units constitute the domestic economy.

- Institutional units
 - not belonging to the domestic economy
 - but has some transactions with resident units
 constitute the Rest of the World (RoW) in the SNA.

42.10 Residence: Criteria

Institutional units	Determined by
Individuals	Residence of the household of which they form part.
Unincorporated enterprises	If not a quasi-corporate, same residence as their owners.
Corporations and NPIs	Normally the country of registration or where legally constituted. Branch in a different country is treated as a quasi-corporate in the host economy.
i. Owners of land, buildings & immovable structures	Deemed always to have a centre of economic interest in the country where they are located. Thus, for all land & buildings owned by non-residents, the SNA assumes a notional resident unit (with non-financial asset and direct investment liability).
ii. Extractors of sub-soil resources.	

42.11 Change of Residence

Individual changing residence from (say) country A to country B:

Changes are recorded only in other changes in volume account.

42.12 State whether TRUE of FALSE.

1. Foreign students staying for three years are considered residents.
2. A branch of Citi Bank (an American bank) in Kolkata is a resident of India.
3. Australian crew of a ship of a Japanese company are residents of Japan.
4. Non-residents are not considered to be owners of immovable assets.
5. All unincorporated businesses without separate accounts belong to household sector
6. Branch of foreign NPI serving residents households is treated as a resident NPISH.
7. Central Bank is part of the general government sector.
8. A branch of a Japanese company in Thailand is resident of Thailand

Q 1. FALSE

Q 2. TRUE

Q 3. FALSE

Q 4. TRUE

Q 5. TRUE

Q 6. TRUE

Q 7. FALSE

Q 8. TRUE

42.13 Exports or Imports

In the SNA, exports (X) and imports (M) [together also called ‘external’ or ‘international’ trade] are such sale-purchase activities which takes place between a resident and a non-resident. Determined by change in economic ownership.

42.14 Exports or Imports or none?

Try to identify which of the following are exports (X) or imports (M) or none (N) of India:

1. Purchase of a Swiss watch by a resident of India from a shop in Kolkata (a city of India).
2. An on-line purchase of UK-published book by a resident of India.
3. Delivery of an Indian monthly magazine subscribed by an Indian student in Oxford University.
4. Payment of hotel charges by a resident of Egypt in Mumbai (a city of India).

N

Since, the purchase is made from a shop, whose owner is also a resident of India.

M

Since, the bookstore in UK is not a resident of India.

N

Since, student in UK is still a resident of India.

X

Since, the hotel is a resident unit of India while the tourist is that of Egypt.

*42.15 Production Process**42.16 Concept of Production*

1993 / 2008 / 2025 SNA:

Production: “ ... an (human) activity in which an enterprise uses inputs to produce outputs (of goods & services).”

Production represents human activities that

- create (produce) goods and / or services
- from inputs of raw materials and other intermediate products,
- using (human) labour and available productive resources.

carried out under the control and responsibility of an institutional unit (an economic entity).

42.17 Production Process

During a period of time, only a part of the capital goods get used up in the process of production in form of depreciation. Also, some non-renewable natural resources are used up in the process in form of depletion.

using

Capital goods

goods & services:

42.18 Few questions

Which of the following acts do you consider as production?

1. Having a cup of tea in the morning
2. Ploughing agricultural field
3. Distribution of newspapers by a delivery boy
4. Repairing a damaged vending machine
5. Natural growth of fish in river
6. Collection of honey from forest

Ans: 1. not production, since no creation is involved.

2. production – an act performed for growing crops

3. production – a trading activity
4. production – adds value by making the vending machine functional
5. not production, does not involve human effort
6. production - collecting honey grown in wild requires human effort

42.19 Factors of Production (contd.)

In macro-economics,

- land (and other natural resources),
- labour,
- capital (financial assets) and
- entrepreneurship

are treated as factors of production, which are directly or indirectly owned by the households.

42.20 Factor Compensation

Services provided by these factors of production are called factor services. The payment made for use of factor services by the enterprises are called factor payment or factor compensation.

Factor of production	Factor payment / compensation
Labour	Compensation of employees (CE)
Land & natural resources	Rent
Capital	Interest
Entrepreneurship	Profit
An indistinguishable mix of labour and other factors	Mixed Income – unincorporated household enterprises' earnings .

Together called

Operating surplus (OS)

42.21 Few more questions

Categorize the roles played by the following into (A) factor services, (B) Intermediate consumption and (C) Capital Goods?

1. Plant & machinery owned by an enterprise
2. An office building hired by a company
3. Wheat used in a flour mill
4. Hired land
5. Share holder of a company
6. Maintenance of owned machinery

Ans: 1. Capital goods, since used for production and owned by the enterprise.

2. Intermediate consumption, since services of the hired building are used by the enterprise and rental paid in exchange.

3. Intermediate consumption, since it is the raw material

4. Factor service

5. Factor service

6. Intermediate consumption. Maintenance cost is treated as intermediate consumption.

42.22 Depreciation / Consumption of Fixed Capital (CFC)

Only a part of the capital goods (and services) is consumed in the production process.

The capital stock undergoes wear and tear.

The value of the wear and tear of the assets used in the process of production is defined as depreciation in the SNA.

Depreciation is the cost of fixed assets used up in production in the accounting period.

It is valued using actual or estimated prices of fixed assets prevailing at the time the production takes place (not the book value).

42.23 Depletion

Much as depreciation, depletion of non-produced natural resources reduces the potential of deriving benefits from these stocks in the future.

Depletion is treated as cost of production in the SNA, as it represents the decline in the value of natural resources due to their extraction in the production of goods.

It represents the value of non-produced natural resources used up in production in the accounting period.

42.24 How are the output of production process used?

The goods and services produced in a production process bring about addition in the stock of produced assets. In the domestic economy, these are used as follows:

All goods and services produced are used in one of these ways.

Products

Final use

Consumption of fixed capital (CFC)

Intermediate consumption (IC)

42.25 What constitute 'Final use'?

Products for final use are either (finally) consumed or forms part of capital formation.

Only a part of capital formation is in fact 'investment'.

This part is called Gross Fixed Capital Formation (GFCF)

Final use

Final consumption

Capital formation

42.26 Gross vs. Net

As a general rule, for all macro-economic aggregates in the SNA, the difference between its 'gross' value and 'net' value is depreciation and depletion (D&D). For example,

$$\text{Net Value Added(NVA)} = \text{GVA minus D\&D}$$

$$\text{Net Domestic Product(NDP)} = \text{GDP minus D\&D}$$

$$\text{Net National Income(NNI)} = \text{GNI minus D\&D}$$

$$\text{Net profit} = \text{Gross profit minus D\&D}$$

$$\text{Net operating surplus} = \text{Gross operating surplus minus D\&D}$$

Net Savings = Gross savings minus D&D

OFFSTAT MIDSEM NOTES

LEC 6

43 Valuation of Domestic Output and External Transactions

43.1 Contents of Session 6

- Output - Definition
- Valuation of domestic output
- Valuation of external transactions

43.2 Output - definition

43.3 Production units and sector

Enterprise is an institutional unit engaged in production. An enterprise may have one or more economic activities. For example, a company that runs a mine and has a number of factories producing different products like steel, machinery and vehicles.

Establishment is a unit which may be an enterprise or part of enterprise. For example, a factory of a company that has more than one factories.

Industry is the aggregate of those establishments that are engaged in same kind of economic activity (according to ISIC).

43.4 A question

How should you distinguish between production and output?

Production is related activities falling within the SNA production boundary.

All that are produced are not included in output.

43.5 Output - Definition

Output is defined (2025 SNA) as the goods and services produced by an establishment, excluding the value of any goods and services

- used in an activity for which the establishment does not assume the risk of using the products in production,
- consumed by the same establishment (except for goods and services used for capital formation or own final consumption).

This definition recognizes that ownership of goods and services used and produced in the production process comes with the associated risks, rewards, rights and responsibilities.

43.6 Inter-establishment flow of output

The following is an example of the most common form of inter-establishment flow of output:

An establishment A (say a farmer) sells products (say wheat) to an establishment B (say a miller) belonging to a different enterprise. B uses it as inputs for its production.

Clearly, the value of sale is output of A and IC of B.

43.7 Valuation of Domestic Output

43.8 Valuation of Goods & Services

The SNA is founded on the premises of equivalence of supply & use of goods & services.

For establishing this equivalence, the SNA recommends a set of standard procedures of valuation for attributing monetary values to economic flows and economic assets.

Why are valuation rules required?

The price for a product paid by a user and that received by its producer are usually very different.

43.9 A Question

A pen purchased from a store in Kolkata for Rs20 . Its producer in Durgapur had received only Rs 10 by selling it.

What do you think are the possible causes for this difference in prices? Which of the following could be the possible reasons for this difference?

- Taxes
- The margins earned by the traders involved
- The transport costs
- All the above

[answer: d.]

43.10 Prices used in valuation

Purchasers' price: at which a purchaser buys a product in the market

Producer's prices: at which the producer sells a product

Basic prices: the price which the producer finally gets by selling a product.

43.11 Valuation Prices

A Consumer buys a pen from a trader at Rs20

The trader purchases each pen at 12 from the manufacturer

The manufacturer pays product tax of 2 to the government.

What is the

purchasers' price of the pen =

trade & transport margin (TTM) =

producer price of the pen =

basic price of the pen =

[Ans: 20 Rs

[Ans: 8 Rs]

[Ans: 20-8=12Rs

[Ans: 12-2=Rs10]

43.12 Relationships – Prices(imp)**

Purchasers' price

Less gross trade and transport margins (and taxes on consumers)

Equals Producer's prices

Less taxes less subsidies on products payable/receivable by their producers

Equals Basic prices.

43.13 Market Prices

In the SNA:

- use of products are recorded at purchasers' prices,
 - output of products are recorded at basic prices.
- since producers' prices are not observable in the expenditures made by the consumers of a product.

The 2008 SNA suggests that the term ‘market prices’ should be avoided when referring to value added. This is because introduction of VAT system taxation may lead to confusion.

While aggregating GVA at the economy level, the 2008 SNA suggests using $GDP = \Sigma GVA_{bp} + \text{product (t-s)}$ instead of using the term ‘GDP at market prices’.

43.14 Valuation of external transactions

43.15 Transactions with the Rest of the world RoW

External trade: exports (X) and imports (M)

Income from and to the RoW: CE and investment income in exchange of factor services

Current transfers: (including income & wealth tax) from and to the RoW

Capital transfers from and to the RoW

Acquisition from and disposal to the RoW of valuables and non-produced assets

Net acquisition and net incurrence of financial assets and liabilities.

43.16 Valuation

Goods valued at f.o.b. (free on board, excluding value of insurance and freight) during change of ownership or as the goods cross the customs boundary.

Usually, at port of entries, customs authorities record the values at c.i.f. (cost, insurance and freight charges).

For national accounts, the value of imports too are recorded in f.o.b.

43.17 Coverage of External Trade

External transaction also includes expenditure on goods and services made in an economy by resident of another economy.

Trade in non-factor services include provision of services to non-residents

- Merchanting of goods
- Service charges for goods received from abroad for processing
- Transport, insurance, hotels and restaurants, etc..

43.18 Merchanting

Purchase of goods by a unit (merchant) in economy X from economy Y and sale to economy Z, where the goods legally change ownership but do not physically enter the economy X.

43.19 External Trade – Exclusions*

- Transactions of non-resident units with RoW
- Flows of sample product, goods for exhibition,
- Flows of goods for repair and for rent
- Flows of goods of returning residents
- Flows of goods for processing
- Factor services (labour, financial resources, etc)
- Goods under operational leasing or for rent

OFFSTAT MIDSEM NOTES

LEC 7

44 Production-side Aggregates

44.1 Contents of Session 7

- Types of output

- Production-side aggregates
- Generation of earned income

44.2 Types of output

44.3 Gross Value of Output (GVO)

Recall that Value Added is defined as value of output minus value of input.

Output of a production process is the goods and services (g&s) produced by it.

In value terms, it is expressed as Gross Value of Output (GVO).

The valuation of output is done in three different ways in the 2008 / 2025 SNA.

44.4 Three-way Distinction of Output*

The 2008 / 2025 SNA makes a three-way distinction of output

- Market output:
- Non-market output:
- Output for own final use:

Those sold in the market at economically significant prices.

Those provided free or at prices that are not economically significant to other institutional units. These are mainly the services produced by the Government, NPISHs and the central bank (in 2025 SNA).

Those used for own final consumption and own-account capital formation.

44.5 Economically significant prices*

Prices are said to be economically significant if they have a significant effect on the amount that producers are willing to supply and the amounts purchasers wish to buy.

The outputs of the general government, the central bank and NPISHs are mostly provided free or at prices that are not economically significant.

44.6 More questions

Which of the following are economically significant prices?

- Price of an entry ticket to a national museum run by the government.
- The price of pen you purchase at your local store.
- The price of a good meal in a restaurant.
- The small amount of registration fees an out patient has to pay in a government hospital providing free health care services.
- Compulsory payments made by financial corporations to the central bank.

[Answers: only b and c]

44.7 Non-Market Output

Non-market output consists of goods and individual or collective services produced by non-profit institutions serving households (NPISHs) or government or the central bank. that are supplied free, or at prices that are not economically significant,

to other institutional units or the community as a whole.

44.8 Output for Own Final Use

Output for own final use consists of products retained by the producing unit for its own use as final consumption or capital formation, such as

- crops produced by a farmer and consumed by his household;
- fixed assets produced by an establishment that are retained; (own-account gross fixed capital forma-

tion);

- changes in inventories of finished goods and work-in-progress intended for one or other of the above uses.
- services provided to households by paid domestic staff;
- imputed services of owner-occupied dwellings;

44.9 A few more questions*

What kind of output are the following?

- a. Services of the police department of a government
- b. Wooden shelves constructed by a retailer for his shop
- c. Housing services provided by owner-occupied dwellings
- d. Housing services provided by rented dwellings
- e. Educational services provided by free schools run by trusts
- f. Services provided by transport corporations.

[Answers: a – non-market output b – for own final use

c – for own final use d – market output

e – non-market output f – market output

44.10 Production-side Aggregates

44.11 Lets begin with a question

A private school in Thailand charges 110 Baht as monthly tuition fees from each of its student. The school has to pay 10% of its earnings as product tax to the government. There were 150 students in a particular month.

What is the GVO at (a) basic prices and (b) producers' prices during the month?

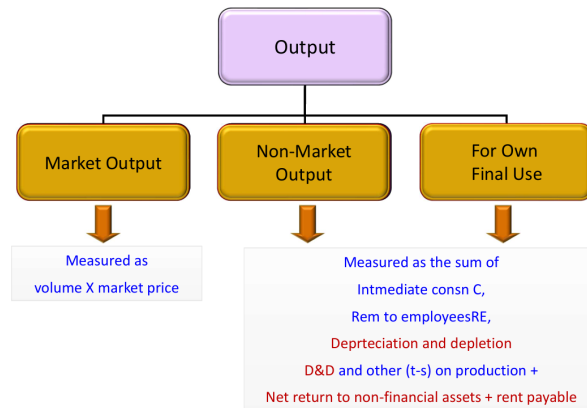
[Answers: (a) 15,000 Baht (b) 16,500 Baht]

44.12 Measuring Non-Market and for-own-use Output

In the recommended approach of 2025 SNA, the gross value of Non-Market and for-own-use Output is to be derived as the sum of

1. Intermediate consumption
2. Remuneration of employees
3. Depreciation (and depletion where relevant),
4. Other taxes (less subsidies) on production other than taxes or subsidies on products,
5. Net return to non-financial assets used in production and
6. Rent payable on the use of non-produced non-financial assets. [In 2008 SNA, rent was restricted to natural resources, which are treated as having infinite life]

44.13 A sum up



44.14 Intermediate Consumption*

Intermediate consumption (IC) is defined as:

Goods and services purchased by producers and

- used as part of production
- consumed during production
- paid for by producer

44.15 GVA at basic prices (GVA_{bp}) of individual units

The GVA_{bp} of corporate sector units and the households

= GVA_{bp} from market output

+ GVA_{bp} from output for own final use.

The GVA_{bp} of the government and the NPISHs (Non-profit institutions serving households)

= GVA_{bp} from non-market output

+ GVA_{bp} from output for own final use.

44.16 GVA_{bp} and GDP of an economy*

Thus, GVA at basic prices of the entire economy

= GVA_{bp} from market output + GVA_{bp} from non-market output + GVA_{bp} from output for own final use.

and the GDP of an economy

= GVA at basic prices of the entire economy

+ Taxes (less subsidies) on products

Or $GDP = GVA_{bp} + (t-s) \text{ on products}$.

44.17 Generation of Earned Income

44.18 Remuneration (Compensation) of Employees

Remuneration (Compensation in 2008 SNA) of employees: is the total remuneration, in cash or in kind, payable by an enterprise to an employee in return for work done by the latter during the accounting period.

Includes:

- wages, salaries, commissions and other benefits to employees
- also social security or provident fund contribution paid by employer for the employees

44.19 Operating Surplus

Operating surplus (OS) : Balance or residual when all the costs are deducted from the value of goods and services produced.

It includes:

- net income (profit) of the producers and
- rent and interest, i.e. the payments made by producers for the use of non-produced non-financial assets and financial assets owned by others.

*44.20 A Question**

During an accounting period, a bread-making unit of a company

- produced bread worth (without taxes): 15,000 Rs
- using flour worth: 10,000 Rs
- paid salary of employees: 1,000 Rs
- and spent on electricity, fuel, & other incidentals: 2,000 Rs

Answers

What is the GVO at basic prices?

What is the value of intermediate consumption?

What is the GVA_{bp}?

What is the operating surplus (OS)?

15,000 Rs

12,000 Rs

(15000-2000-1000)

3,000 Rs

2,000 Rs

OFFSTAT MIDSEM NOTES

LEC 8

45 Income-side Aggregates

45.1 Contents of Session 8

- Earned (Primary) income & its distribution
- Balance of Earned (Primary) income / GNI
- Use of disposable income

*45.2 Earned (Primary) Income & its distribution**45.3 Households' earnings and Earned (Primary) Income*

Two broad categories of household earnings:

1. from participation or involvement in a production process
2. others

The earnings of the first category are called earned (primary) income. Earned (Primary) incomes include those earned from the other institutional units as well.

Earned Income which includes earnings of

- wages & salaries,
- rent from leased-out non-produced non-financial assets,
- dividends and

- Interests on deposits from banks / funds lend to enterprises.

include all kinds of transfer receipts, receipts from sale of property, earnings from holding of assets (holding gains/ loss) etc.

45.4 *Earned (Primary) Income - Definition*

The income generated in a production process is the GVA at producers' prices of the production unit.

It is distributed as earned (primary) income to those providing factor services as factor compensations and to the government as production taxes (less subsidies).

Definition of earned (primary) income:

The income of the recipients of the shares of income generated in production processes is called earned (primary) income in the SNA.

45.5 *Earned (Primary) Income - from recipients' perspective*

Households and financial institutions receive primary income from enterprises as

- Remuneration / Compensation of employees (RE / CE) and
- Property income (PI) from
 - lending of financial assets : interest and dividends
 - renting of natural resources: rent
- Mixed income (MI): a mix of RE/ CE & PI

while the government receives

- (Taxes less subsidies) on production and imports

[(t-s) on production includes both (t-s) on products as well as other production (t-s)]

45.6 *Which of the following (more than one) are earned (primary) income, i.e. income obtained from production process?*

1. Beggars' income
2. Smugglers Income
3. Gamblers' income
4. Income from purchase and sale of shares
5. Landlord's income from rent of land
6. Product taxes
7. Receipts from sale of land & existing building
8. Government's revenue from income tax

Answer: Only (b), (e), and (f) are earned income.

(b) is a trading activity though illegal; (e) is a factor income and (f) part of income generated from production process.

(a), (c) and (h) are transfers and (d) is holding gain / loss. These and (g) do not arise from any current-period production process.

45.7 *Property Income*

Includes:

- Investment income
 - Interest: nominal interest receivable or payable + / – allocated IFSLD (called FISIM in 2008 SNA)
 - Distributed income of corporations like dividends.
 - Other investment income like
 - * attributable to insurance policyholders,

* payable on pension entitlements, etc.

- Rent: Income receivable by the owner of a natural resources.

[FISIM: Financial intermediation services indirectly measured

IFSLD: implicit financial services on loans and deposits]

45.8 Property Income and Operating Surplus

Operating surplus gets distributed as property income:

- Rent
 - Goes to owner of land & other non-produced natural assets
- Investment income
 - Is paid to the provider of financial assets

45.9 Property Income and Operating Surplus

The payments and receipts of property income (PI) gets cancelled out within the domestic economy.

What the households finally get is the operating surplus generated in the production undertaken within the domestic economy.

Thus, property income (PI) does not appear in the identity for GDP by income approach

$$GDP = \text{Remuneration of employees (RE)} + \text{Operating Surplus (S)} \\ + \text{Mixed Income (MI + product (t-s) + (t-s) on import}$$

t: tax ,s: subsidy

The RE in this identity represents the compensation of employees received from resident enterprises only.

45.10 Balance of Earned (Primary) income / GNI

45.11 Flow of Earned (Primary) Income of a household

During an accounting period, a household, with a member working in a factory and another running a grocery, made the following transactions:

45.12 National Income

1. Salary received from the factory: 5,000
2. Purchased processed food for sale from the same factory: 10,000
3. Remittances received from a relative living abroad: 5,000
4. Paid contribution to Traders' Association (NPISHs): 200
5. Paid income tax of 10% on salary earnings: 500
6. Sells processed food: 15,000
7. Purchased household provisions: 4,350
8. Paid for electricity consumed in the grocery: 150
9. Paid to hired worker: 1,000
10. Paid sales (product) tax on sale of processed food 150

45.13 A Question

What are the values of earned (primary) income in-flow to and out-flow from the household?

45.14 National Income

1. Salary received from the factory: 5,000
2. Purchased processed food for sale from the same factory: 10,000
3. Remittances received from a relative living abroad: 5,000

4. Paid contribution to Traders' Association (NPISHs): 200
5. Paid income tax of 10% on salary earnings: 500
6. Sells processed food: 15,000
7. Purchased household provisions: 4,350
8. Paid for electricity consumed in the grocery: 150
9. Paid to hired worker: 1,000
10. Paid sales (product) tax on sale of processed food 150

Answer: Earned income in-flow = salary received from the factory + GVA (income) from the grocery =
 $= 5000 + (15000 - 10000 - 150) = 9850$

Earned income out-flow = salary paid + sales tax = $1000 + 150 = 1150$.

45.15 Balance of Earned (Primary) Income

For an institutional unit,

Balance of earned (primary) Income is

total value of the earned (primary) income receivable less total of the earned (primary) incomes payable.

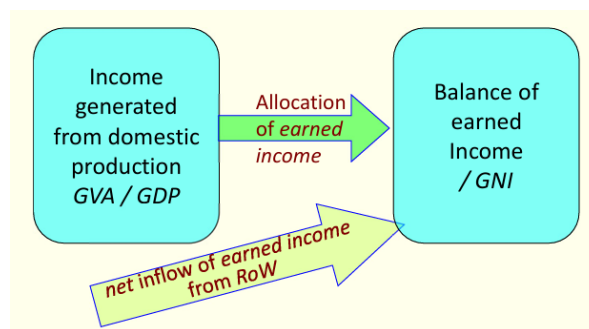
For an institutional sector or a segment of an economy, the balance of earned (primary) income can be similarly worked out.

45.16 Earned (Primary) Income to & from RoW

Residents of an economy receive earned (primary) income from the non-residents for providing factors services for production activities carried out in the ROW. Conversely, the resident enterprises pay earned (primary) income to non-residents for using factor services provided by them.

45.17 Flow of Earned (Primary) Income from and to RoW

The residents of an economy receive primary income from two distinct sources: domestic economy and RoW



At the level of the total economy,

the sum of Balance of Earned (Primary) Income over all residents is called Gross National Income, GNI.

45.18 Gross National Income (GNI)

Gross National income (GNI) of an economy represents the income of its residents.

- GDP measures the total production carried out within the economic territory of a country,
- GNI measures the total income of all economic agents residing in the territory.

To arrive at an estimate of GNI, the GDP has to be adjusted for the net earned (primary) income of the residents from the RoW.

Thus,

$$\text{Gross National Income (GNI)} = \text{GDP} + (\text{net}) \text{ earned income earned by residents from RoW.}$$

45.19 Few questions

Which of the following is true about Gross national income (GNI)?

1. includes income tax from the rest of the world (RoW) received by the government
2. is the measure of production of an economy
3. is equal to GDP plus (net) earned (primary) income (of the residents) from RoW
4. includes money received by resident households from a non-resident relative
5. is the measure of earned income of the residents of an economy
6. includes receipts from consultancy services provided to the rest of the world (RoW)

Answers: Only (c), (e) and (f) are true.

*45.20 Flow of transfers and disposable income**45.21 Transfers*

Transfers are unilateral transactions in which one economic entity provides a real resource, such as goods or services, or a financial item, to another entity without receiving any real resource or financial item in exchange.

Current transfers (other than social transfers in kind)

- Current taxes on income, wealth, etc.
- (Net) social contributions and social benefits
- Other current transfers

45.22 Social contributions

Social contributions are current transfers that are payable by the households (actual or imputed) to social insurance schemes to make provision for social insurance benefits to be paid.

- Social contributions may be made by employers on behalf of their employees.

[Where is this included in generation of earned income?]

- Mostly, social-contributions-receiving units fall in the general government sector and insurance corporations & pension funds in the financial corporation sector.
- Social contributions are payable only by households, either resident or non-resident.

45.23 Social benefits (other than those in kind)

Social benefits are current transfers receivable by households and payable by social insurance schemes, such as provident funds, life insurance and health insurance schemes.

- These become receivable only when certain events or circumstances arise.
- These arise from, for example, sickness, unemployment, retirement, housing, education or family circumstances.
- Social benefits may be provided under social insurance schemes or by social assistance.

45.24 Other current transfers

Other current transfers consist of all current transfers between resident institutional units, or between resident and non-resident units, other than

- current taxes on income, wealth, etc.,
- social contributions and benefits, and
- social benefits in kind.

45.25 Other current transfers

This kind of current transfers includes:

- net premiums and claims under non-life insurance policies,

- current transfers between different kinds of government units, including those between general government and foreign governments,
- current transfers to and from NPISHs
- between residents and
- between resident and non-resident households.

45.26 Insurance premiums

- Insurance premiums are not entirely current transfers.
- These include an element of payment for services provided by the insurance agency.
- The benefits, however, is considered to be purely current transfers.

45.27 Which of the following are current transfers?

1. Payment of corporate tax.
2. Donation for construction of school building.
3. Payment of life-insurance premium.
4. Payment made for election bonds
5. Payment of non-life-insurance premium.
6. Money sent home by a non-resident.
7. Receipts of insurance claims by a corporate sector unit.
8. Receipts of royalty for use of a patented design.
9. Contribution made by a worker to the labour union of which she is a member.

Ans.: entirely – 1, 4, 6, 7 & 9 partly – 3 & 5

45.28 Disposable Income

Disposable income represents the purchasing power of the residents.

It is derived from the balance of earned incomes of an institutional unit or sector by:

1. adding all current transfers, except social transfers in kind, receivable by that unit or sector; and
2. subtracting all current transfers, except social transfers in kind, payable by that unit or sector.

For a unit or an institutional sector of the economy,

$$\text{Disposable income} = \text{Balance of earned income} + (\text{net}) \text{ current transfer receivable.}$$

45.29 Disposable Income (2)

- Besides the cross-border flow of earned income, there are other cross-border transactions that determine the purchasing power of the residents.
- The current transfers from or to the RoW makes the purchasing power at the disposal of the residents different from the GNI.
- The resulting disposable income of the residents is called gross national disposable income (GNDI) and is defined as

$$= \text{GNI} + (\text{net}) \text{ current transfers receivable from the RoW.}$$

45.30 Another Question

What are the values of transfers in-flow to and out-flow from the household? What is its disposable income?

1. Salary received from the factory: 5,000
2. Purchased processed food for sale from the factory: 10,000
3. Remittances received from a relative living abroad: 5,000

4. Paid contribution to Traders' Association (NPISHs): 200
5. Paid income tax of 10% on salary earnings: 500
6. Sells processed food: 15,000
7. Purchased household provisions: 4,350
8. Paid for electricity consumed in the grocery: 150
9. Paid to hired worker: 1,000
10. Paid sales (product) tax on sale of processed food 150

45.31 Another Question: Answers

Answer: the disposable income of the household is as follows:

Primary income: $a + (f - b - i - j) = 5000 + (15000 - 10000 - 1000 - 150 - 150) = 8700$

- We also know that the household had received / paid
 - remittances from a relative living abroad (5000)
 - contribution to Traders' Association (200) and
 - income tax of 10% on salary earnings (500).
- Thus, the net transfers receivable of the household = $c - d - e = 5000 - 200 - 500 = 4300$ and
 - its disposable income = earned (primary) income + net transfers receivable $= 8700 + 4300 = 13000$.

45.32 Gross national disposable income (GNDI)

This is related to GDP and GNI in the following way:

$$GNDI = GNI + (\text{net}) \text{ current transfers from RoW}$$

$$= GDP + (\text{net}) \text{ earned income from RoW}$$

$$+ (\text{net}) \text{ current transfers from RoW.}$$

45.33 Use of Disposable Income

45.34 Consumption

Consumption of goods and services is the act of using them.

The 2008 SNA recognizes three distinct kinds of consumption

- Intermediate consumption
- Depreciation (CFC)
- Final consumption

The consumption of fixed capital (CFC) is called “depreciation” in 2008 SNA.

45.35 Consumption

- Consumption of goods and services is the act of completely or partially using them up
 - completely in a process of production (intermediate consumption) or
 - partially in a process of production (depreciation) or
 - for direct satisfaction of human needs / wants (final consumption)
- The third activity represents final consumption and includes use of goods and services for both individual needs and collective human needs.

45.36 Final Consumption

Goods & services are finally consumed by the households.

Final consumption includes use of goods and services for both individual needs and collective needs.

- individual needs - acquired by a household and used to satisfy the needs or wants its members; and
- collective human needs - service provided simultaneously to all members of the community or a section of the community.

These are provided only by the general government and the central bank.

45.37 A question

During an accounting period, a Malaysian farmer's household has

- mixed income from farming = 5000 Ringgits
- earnings from rent = 1000 Ringgits
- retains grains produced
 - for use as seeds in next year = 100 Ringgits
 - for own consumption = 900 Ringgits
- purchases g&s for consumption = 4000 Ringgits
- purchases fertilizers for farming = 500 Ringgits

What is the final consumption expenditure (FCE) of the household?

[Ans: $900 + 4000 = 4900$]

Only the grains retained for own consumption and purchased g&s for consumption are included in FCE]

45.38 Final Consumption Expenditure (FCE)

FCE is incurred by the households, government and the Non Profit Instns serving households(NPISHs).

[In the SNA, FCE is NOT admissible for corporate sector units]

Thus the final consumption expenditure of an economy is the sum of

- Household final consumption expenditure (HFCE)
- Government final consumption expenditure (GFCE)
- Central bank's final consumption expenditure (CBCE)
- Final consumption expenditure of the NPISHs

Often the HFCE and the final consumption expenditure of the NPISHs are together called "Private Final consumption expenditure (PFCE)".

Thus,

$$FCE = GFCE + CBCE + PFCE$$

45.39 Household Final Consumption Expenditure (HFCE)

HFCE includes

- Consumption expenditure of resident households in the economic territory
 - from domestic production
 - from imports of goods and services
- Consumption expenditure of resident households in other economic territory
 - Considered as imports of goods and services

45.40 Government Final Consumption Expenditure (GFCE), CBCE and NPISHs' Final Consumption Expenditure

The value of the entire non-market output produced by the general government, the central bank and the NPISHs is included in their respective final consumption expenditure.

Though in most cases households make the final consumption of the non-market output produced by the general government and the NPISHs, the expenditure is NOT incurred by them.

45.41 Government Final Consumption Expenditure (GFCE)

Also includes government purchases intended primarily as current transfer (or benefits) in kind, such as

- Government purchase of food to be distributed to the victims of flood
- Government purchase of blankets to be distributed to the homeless

$$GFCE = \text{Total Output of Government Sector}$$

$$- \text{Own account gross capital formation}$$

$$- \text{Sales of goods and services}$$

$$+ \text{Benefits in kind}$$

(purchases of market goods and services to be provided free to households)

45.42 Another question

During an accounting period, a Malaysian farmer's household has

- mixed income from farming = 5000 Ringgits
- earnings from rent = 1000 Ringgits
- receivable current transfer = 2000 Ringgits
- final consumption expenditure = 4900 Ringgits

Assuming CFC of 500 Ringgits, what is the gross savings and net savings of the farmer?

[Ans: Disposable income = 5000 + 1000 + 2000 = 8000

FCE = 4900

Thus, gross savings = 8000 - 4900 = 3100

and net savings = 3100 - 500 = 2600]

45.43 Savings

Thus, savings is defined as

$$\text{gross savings} = \text{GNDI (Gross National Disposable Income)} - \text{final consumption expenditure}$$

OR

$$\text{net savings} = \text{GNDI} - \text{final consumption expenditure} - \text{D\&D}$$

OFFSTAT MIDSEM NOTES

LEC 9

46 Capital Formation and Net Lending / Borrowing

46.1 Contents of Session 9

- Asset Classification
- Capital Formation
- Net lending/ borrowing
- Main Identities

46.2 Asset Classification

46.3 Classification of Assets – Financial assets

Financial assets: These are assets that originate in the contractual relationship between two institutional units whereby one party acquires the right to receive a payment, or series of payments, from the other party.

For example: currency, shares and bonds, deposits in banks etc.

46.4 Financial Liabilities

- **What are liabilities?**
 - Is established when one unit (the debtor is obliged to provide a payment to another unit (the creditor)
- **What is a financial claim?**
 - A financial claim is the payment due to the creditor by the debtor under the terms of a liability
 - The claim is extinguished when the liability is discharged by the debtor paying a sum agreed in the contract

46.5 Classification of Non-financial Assets

Broad groups:

- **Produced assets** (other than natural resources)
 - Fixed assets
 - Inventories
 - Valuables
- **Non-produced assets** (other than natural resources)
 - Contracts, leases and licenses
 - Goodwill & marketing assets
- **Natural resources (assets)**
 - Produced biological resources
 - Non-produced assets

46.6 Classification of Assets – non-produced assets

Non-produced assets (other than natural resources): non-financial assets that come into existence other than through production processes.

Examples:

- contracts, leases and licenses, and
- purchased goodwill and marketing assets.

46.7 Produced assets (other than produced natural resources)

Produced assets: non-financial assets (other than natural resources) that come into existence through the production process.

They consist of

- **fixed assets** – used repeatedly in the process of production for more than one year
- **inventories** – goods or services that are held for sale, use in production, or for use at a later date
- **valuables** – Assets held primarily as store of value

46.8 Fixed assets

Fixed assets are used repeatedly or continuously in the process of production for more than one year.

Fixed assets include:

- structures,
- machinery and equipment
- intellectual property products [since 2008 SNA]

Excludes cultivated assets such as trees or animals that are used repeatedly or continuously to produce other products.

46.9 Classification of Fixed Assets

Fixed assets by type of asset:

- Dwellings
- Other buildings and structures
- Machinery and equipment
- Weapons systems
- Costs of ownership transfer on non-produced assets
- Intellectual property products

Land improvements introduced in 2008 SNA.

Weapons systems introduced in 2008 SNA.

Intellectual property products introduced in 2008 SNA.

46.10 Classification of Fixed Assets (contd.)

Intellectual property products

- Research and development
- Mineral exploration and evaluation
- Computer software
- Data & Databases
- Entertainment, literary or artistic originals
- Other intellectual property products

46.11 Classification of Natural resources

- Land
- Mineral
- Energy resources – both renewable and non-renewable
- Water resources
- Non-cultivated biological resources
- Cultivated biological resources
 - Animal resources yielding repeat products
 - Tree, crop and plant resources yielding repeat products
 - Animal resources yielding once-only products
 - Tree, crop and plant resources yielding once-only products
- Others such as radio spectra. These can be both produced and non-produced assets.

46.12 Inventories

Goods or services that are held for sale, use in production, or for use at a later date.

These consist of:

- Materials and supplies [raw materials etc.]
- Work-in-progress [example: unfinished products]
- Finished goods [produced but not yet sold]
- Military inventories [like ammunitions]
- Goods for resale [i.e. traders stocks of goods for sale]

46.13 Valuables

Assets held primarily as store of value.

Valuables are neither used for consumption – final or intermediate – nor as fixed assets for further production.

Commonly held valuables are:

- precious metals and stones,
- antiques and other art objects, and
- other valuables (e.g., jewelry of significant value)

Acquisition minus disposal of these is an expenditure-side aggregate.

46.14 Capital Formation

46.15 Gross Domestic Capital Formation (GDCF)

Gross Domestic Capital Formation, GCF is acquisition less disposal of produced assets for purposes of (gross) fixed capital formation, inventories or valuables.

GDCF consists of:

- Gross fixed capital formation (GFCF)
- Changes in inventories (CII)
- Acquisitions less disposals of valuables

46.16 Gross Fixed Capital Formation (GFCF)

GFCF is the addition (acquisition less disposal) of (human-made) produced assets that are used repeatedly for production, including produced assets acquired by purchases, barter, capital transfers and produced for own use.

It includes:

- Acquisition less disposal of new or existing produced assets
- Costs of ownership transfers on non-produced, non-financial assets
- Major improvements to produced and non-produced, non-financial assets that extend the lives of assets

46.17 Change in Inventories (CII)

Enterprises have stocks of these at the beginning and end of an accounting period.

In National Accounts, the difference between the stocks at the end and at the beginning of the accounting period is called “Change in Inventories”.

Inventories include:

- unsold finished products
- goods for sale with the traders
- unused raw materials (inventories)

- work-in-progress

46.18 Net lending / borrowing

46.19 Savings and Investment in an Economy

- GDCF or GCF represents investment of an economy.
- But it is not financed entirely by gross savings.
- In addition to the cross-border current transfers, there is another kind of transfers called capital transfers.

46.20 Capital Transfers

Capital transfers are those that transfer ownership of a fixed (capital) asset or are linked to acquisition or disposal of a fixed asset or involve forgiveness of a liability (say bad debt) by a creditor.

Example:

- Grant for construction of school building / religious institution
- Waiving of debts
- Capital taxes

46.21 A question

A non-profit institution (NPI), providing free education to street children, receives a donation of Rs. 10,000 in an accounting period.

It pays salary & wages of Rs. 7000 to the teaching and non-teaching staff. It pays no tax and its CFC is assumed to be Rs. 200.

For construction of a class room, the NPI received a capital grant of Rs. 50,000. But, construction of the class room requires Rs. 100,000.

1. What is the savings of the NPI?
2. How much fund does the NPI have for construction of the class room?
3. How can the NPI meet the shortfall?

Ans.:

1. Rs. 10,000 – Rs. 7,000 – Rs. 200 = Rs. 2,800
2. Rs. 2,800 + Rs. 50,000 = 52,800
3. The NPI has to borrow (Rs. 100,000 – Rs. 52,800) Rs. 47,200

46.22 Financing capital formation

- Capital transfers together with gross savings is taken as the resources for investments (or GDCF) at the disposal of the residents.
- But, in a given accounting period, the resources available for capital formation are often not enough to finance GDCF.

46.23 Savings and Investment

Investments include Savings, capital transfers and Borrowings

46.24 Net lending and borrowing

The borrowings and lending are made by acquiring or selling

- shares of foreign companies or
- providing business loans or
- saved in foreign banks or
- by acquiring financial assets of other forms.

In the SNA, all these kinds of investments in the RoW are treated as acquiring of financial assets, which

in fact represent lending to RoW.

46.25 Net lending and borrowing (contd.)

Thus,

$$\text{Net lending / borrowing from /to RoW} = \text{Gross Savings} + (\text{net}) \text{ capital transfer from RoW} - \text{GDCF}.$$

46.26 Basic Framework of SNA

46.27 Relook at GDP by Expenditure approach*

The value of production of the domestic enterprises (GDP) is realized as aggregate of the monetary flows from

- households as HFCE (household final consumption expenditure)
- government as GFCE (Govt final consumption expenditure), central bank (CBCE) and NPISHs
- other domestic enterprises on account of purchase of capital goods & services (GDCF) and
- rest of the world (RoW), on account of exports (X) net imports (M).

$$GDP = HFCE + CBCE + FCE \text{ of NPISHs} + GFCE + GDCF + X - M$$

Including the expenditure of the NPISHs in HFCE,

$$GDP = PFCE + GFCE + CBCE + GDCF + X - M$$

46.28 Three approaches of deriving GDP

The fundamental macro-economic equations:

$$\text{Production} = GDP$$

$$= GDP_{bp} + (t - s)_{products}$$

$$\text{Income} = RE + OS + MI + \text{product} (t - s)$$

$$\text{Expenditure} = PFCE + CBCE + GFCE + GDCF + X - M$$

46.29 Main identities

The following is a summary presentation of the main identities of the SNA.

These identities form the basis of the sequence of transaction accounts of the system.

46.30 Main Identities (Contd.)

Commodity balance:

$$GVO \equiv IC + PFCE + GFCE + CBCE + GFCE + CII$$

$$+ \text{acquisition less disposal of valuables} + X - M \quad \dots [1]$$

where PFCE stands for private final consumption expenditure, which is HFCE and final consumption

expenditure of the NPISHs.

IC: intermediate consumption, CII: change in inventory

GFCE: Govt final consumption expenditure

CBCE: Central Bank CE

GFCF: Gross Fixed Capital Formation

X-M : Export-Import

Production-side identity:

$$GDP \equiv GVO_{bp} - IC + \text{product } (t - s) + (t - s) \text{ on imports} \quad \dots [2]$$

Gross Domestic Product

Tax and subsidy

46.31 Main Identities (Contd.)

Income-side identities:

RE: Remuneration of employees, OS: Operating surplus, MI: Mixed Income, PIROW: Primary Income from rest of the world, t-s: tax-subsidy, GDP: Gross Domestic Product, GNI: Gross National Income, GNDI: Goss National Disposable Income

$$GDP \equiv (RE + OS \& MI) \text{ generated in domestic enterprises}$$

$$+ \text{product } (t - s) + (t - s) \text{ on imports} \quad \dots [3]$$

$$GNI \equiv (RE + OS \& MI) \text{ generated in domestic enterprises}$$

$$+ \text{product } (t - s) + (t - s) \text{ on imports}$$

$$+ RE \text{ from RoW (net)} + PI \text{ from RoW (net)} \quad \dots [4]$$

$$GNDI \equiv GNI + (\text{net}) \text{ current transfers}$$

$$+ (\text{Net}) \text{ taxes on income \& wealth from RoW} \quad \dots [5]$$

46.32 Main Identities (Contd.)

Expenditure-side identities:

PFCE: private final consumption expenditure, GFCE: Govt final consumption expenditure, CBCE: Central Bank CE

GFCF: Gross Fixed Capital Formation, CII: change in inventory

GNDI: Goss National Disposable Income

$$GDP \equiv PFCE + GFCE + CBCE + GFCF + CII$$

$$+ \text{acquisition less disposal of valuables} + X - M \quad \dots [6]$$

$$\text{Gross Savings} \equiv GNDI - PFCE - GFCE - CBCE \quad \dots [7]$$

This leads to the final identity relating to transactions.

46.33 Main Identities (Contd.)

$$\begin{aligned} \text{Net lending / borrowing from /to RoW} &\equiv \text{Gross Savings} + (\text{net}) \text{ Capital transfer receivable} \\ &\quad - (\text{GFCF} + \text{CII} + \text{acquisition less disposal of valuables}) \\ &\quad - \text{acquisition less disposal of non-produced non-financial assets} \\ &\quad \dots [8] \end{aligned}$$

46.34 Main SNA Aggregates – A sum up*

$$\begin{aligned} &\text{Gross Domestic Product (GDP)} \\ &+ (\text{net}) \text{ earned income from RoW (Rest of the world)} \\ &= \text{GNI (Gross National Income)} \\ &+ (\text{net}) \text{ current transfer from RoW} \\ &+ (\text{net}) \text{ taxes on income \& wealth from RoW} \\ &= \text{GNDI (Gross National Disposable income)} \\ &\quad - \text{final consumption expenditure} \\ &\quad = \text{Gross Savings} \\ &\quad + (\text{net}) \text{ capital transfer from RoW} \\ &\quad - \text{gross capital formation} \\ &\quad - \text{acquisition less disposal of valuables from RoW} \\ &\quad - \text{D\&D} \\ &= \text{Net lending to/ borrowing from RoW} \end{aligned}$$

OFFSTAT MIDSEM NOTES

LEC 10

47 SNA Sequence of Economic Accounts – An Overview

47.1 Sequence of Economic Accounts – An Overview

47.2 The accounts

The accounts of the SNA summarises all economic flows and stocks.

These are divided into two main classes:

- The sequence of economic accounts, which includes the full sequence for
 - each institutional sector
 - the RoW and
 - total economy
- The other parts of the integrated framework, which includes
 - Supply and use framework – Input-output tables
 - Labour market tables
 - Capital services

47.3 Full sequence of economic accounts - Characteristic features

The SNA sequence of economic accounts provides a comprehensive record of an economy's production, income, consumption, accumulation and wealth.

The individual accounts are classified into three categories:

1. Current accounts
2. Accumulation accounts and
3. Balance sheets.

Each account in the sequence contains one or more aggregates, each of which has a particular economic interpretation.

The consistency between individual accounts is ensured by the use a single production boundary and a single assets boundary.

47.4 Net Worth – Wealth of an Economy

Net worth – a measure of wealth of an economy - is defined as

$$\text{Value of total assets minus total liabilities}$$

Assets include only the economic assets, and are of three broad kinds, viz.

- produced assets
- non-produced assets – natural and other assets and
- financial assets

Liabilities are all financial.

The **Balance sheets** of the SNA sequence of accounts gives a summary description of the wealth of an economy.

47.5 Balance Sheets in the SNA

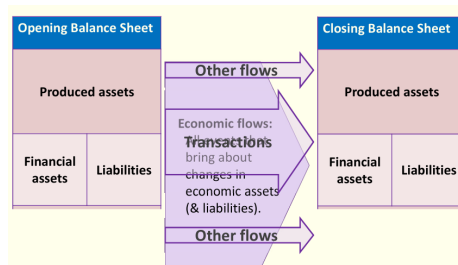
The SNA sequence of accounts includes two balance sheets for an accounting period

- **Opening Balance Sheet:** depicts the position at the beginning of the accounting period
- **Closing Balance Sheet:** depicts the position at the end of the accounting period.

The balance sheets include **net worth** as the balancing item between the assets and liabilities.

The rest of the SNA sequence of accounts capture the **economic flows** that bring about the changes in the balance sheets.

47.6 Accounting Framework [for an economy]



47.7 Transactions in Sequence of accounts

In the sequence of accounts, all transactions are captured in a series of separate accounts, which are called 'Transaction Accounts'.

Two kinds of transactions:

- Those which **do not affect the balance sheet entries directly**:
 - Production
 - Consumption
 - Flows of income etc.
- Those which **lead to changes in the balance sheet entries**:
 - Capital formation
 - Transactions in non-produced natural assets
 - Transactions in financial assets & liabilities

These are recorded in current accounts.

These are recorded in Accumulation of assets accounts.

47.8 Transaction accounts

Production account – all production-related transactions,

Income accounts - transactions relating to earned income generation & distribution, transfers, consumption

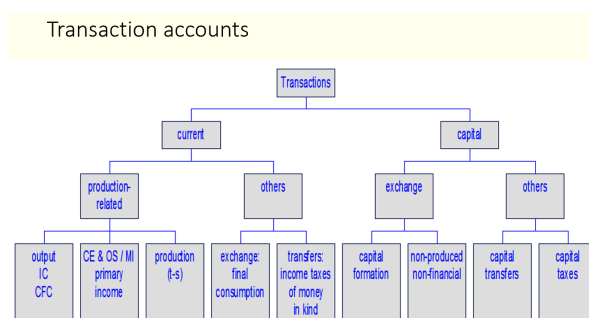
These are the accumulation of assets accounts

Capital account - capital formation & capital transfers

Financial account – all transactions involving money flow

These are the current accounts

47.9 Transaction accounts



47.10 General Features of Accounts

Entries made in these accounts are based on the principle of **double accounting**, thus permit checking consistency.

Like business accounts, each of the accounts of SNA is organised as a T-accounts.

47.11 Two sides of Transaction Accounts

Each transaction is recorded twice on two sides.

Two sides are called

- ‘revenues’ and ‘expenditures’ for **current accounts**
- ‘changes in liability & net worth’ and ‘changes in assets’ for **accumulation accounts**
- ‘liabilities & net worth’ and ‘assets’ for **Balance sheet**

Total of transactions on the two sides must be equal.

47.12 Applicability of Accounting structure

The entire accounting structure can be applied to record all transactions and other flows of a

- single institutional unit or

- an institutional sub-sector / sectors as well as
- total economy.

However, all transactions are not relevant for all sectors.

The SNA recommends compilation of the sequence of accounts separately for each of the five institutional sectors.

47.13 *Balancing Items*

The balancing item of an account

- is the sum of resources (right side) minus the sum of uses (left side) in each of the current accounts,
- and is shown on the left side (uses side) of the account.

47.14 *Balancing Items (contd.)*

The balancing item of each account is significant as a macro-economic aggregate like

- gross / net domestic product (GDP / NDP)
- gross / net national income (GNI/ NNI)
- disposable income
- saving and
- net lending/borrowing.

47.15 *Links between Accounts*

Among the transaction accounts, the **balancing item** (in the uses side) of one account is carried forward as the first item (in the resources) of the next account.

The changes in assets and liabilities brought about by transactions (and other changes) are reflected in the Balance Sheet.

The sequence of accounts thus provides an integrated view of the entire economy.

47.16 *Structure of the Income Accounts*

- Generation of earned income account
- Allocation of earned income account
 - Entrepreneurial income account
 - Allocation of other earned income account
- Income transfers other than social transfers in kind account
 - Social transfers in kind account
- Use of income
 - Use of disposable income account
 - Use of disposable income adjusted for social transfers in kind account

47.17 *Accumulation of Assets Accounts*

Among the transaction accounts, the **Capital** and **Financial** Accounts are the **Accumulation accounts** of the system.

These accounts record transactions in balance-sheet items, i.e., in assets and liabilities

RHS: “Changes in liabilities and net worth”, instead of “Revenues”

LHS: “Changes in assets”, instead of “Expenditures”.

47.18 *Accumulation of Assets Accounts (contd.)*

Both Accumulation accounts and Balance sheets are concerned with ownership.

Balance sheets specify the ownership of assets and liabilities to other units at given points in time,

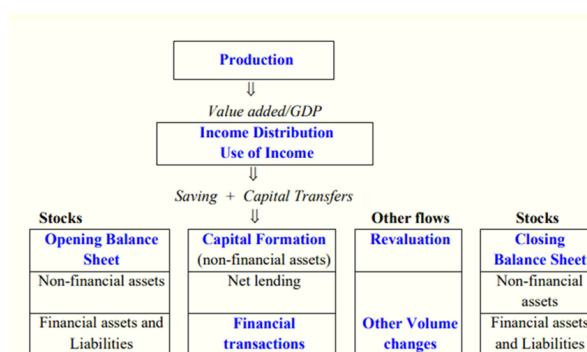
Accumulation accounts show the changes in assets and liabilities over given periods.

In this course, the accounts for recording the other flows are not included.

47.19 Balance Sheet

- Shows the stock of assets on the left-hand side and financial liabilities & net worth on the right-hand side of the account
- A balance sheet may be drawn up for
 - institutional units
 - institutional sectors and
 - the total economy

47.20 Sequence of Accounts– An Overview



OFFSTAT MIDSEM NOTES

LEC 11

48 Monetary Transaction Account

48.1 Contents

- Transactions from Compilers' perspective
- Money Balance Identity
- Monetary Transactions Account (MTA)
- Aggregating MTA

48.2 Transactions from Compilers' perspective

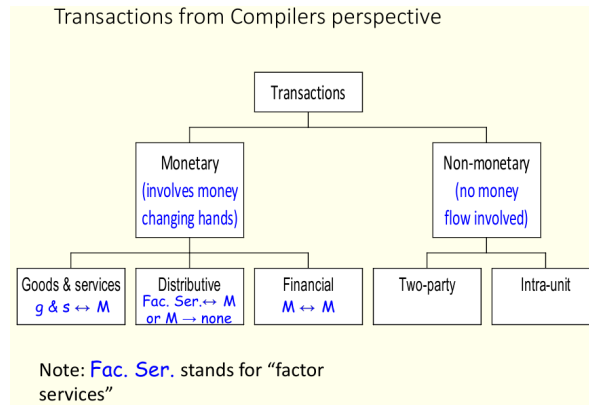
48.3 Forms of transactions

Transaction takes place in different forms. Their values are obtained in different ways. For example,

1. Most commonly, a transaction takes place between two parties. Generally, their values are observable.
2. There are transactions involving only one unit – intra-unit transactions. Their values are not observable.
3. The values of transactions in non-produced goods and services (that come into existence other than through production processes). are not always observable.
4. Often, a single financial transaction represents two more analytically different transactions.

The values of transactions of (b), (c) & (d) kinds are either imputed or derived indirectly by the national accountants.

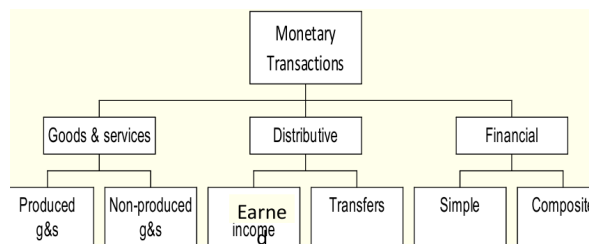
48.4 Transactions from Compilers perspective



48.5 Monetary Transactions

– those involving money changing hands

All the monetary transactions, except the ‘composite financial’ ones, are recorded as they are observed. The ‘composite financial’ transactions are composed of two or more, either analytically same or different, kinds of transactions.



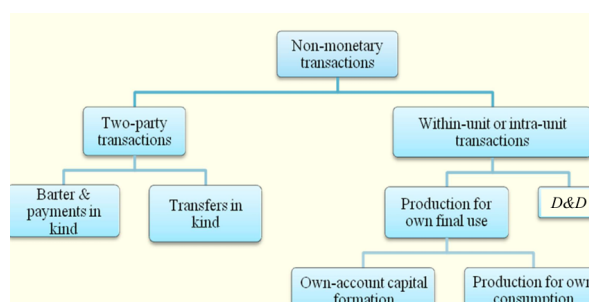
48.6 Monetary transactions

The SNA defines:

A monetary transaction is one in which one institutional unit makes a payment (or receives a payment) or incurs a liability (or receives an asset) stated in units of currency in exchange for goods and (factor and non-factor) services.

- All monetary transactions are between two units. These are mostly observable.
- Composite financial transactions are not recorded as they are observed, e.g.
 - nominal interest, total insurance premiums etc.
- These are not recorded as they are observed.
- In the SNA, they are split up into various components and their classification and routing are modified.

48.7 Non-Monetary Transactions



48.8 Non-monetary transactions

- Non-monetary transactions are not observable.
- The values of such transactions are imputed by the national accountants.
- These are either two-party or intra-unit.
 - The two-party non-monetary transactions are mostly treated the same way as the monetary transactions.
 - The SNA recognises some notional flows as intra-unit or internal transactions for analytically useful presentation.

*48.9 Money Balance Identity**48.10 A Simple Illustration*

Take the case of a person, who in the beginning of the day, had 100 in her wallet.

During the day, she

- spends 20 for her lunch
- pays 30 for transport to work place
- receives a payment of 150 from her client
- takes a loan of 350 from a friend and
- purchases stationery of 500 for her office.

How much will she have left in her wallet at the end of the day?

Obviously, she will have 50 ($= 100 - 20 - 30 + 150 + 350 - 500$) in her wallet at end of the day.

48.11 Illustration (contd.)

Note that, during the day, there was a money in-flow of 500,

150 (receipt of payment from client) and

350 (from the loan taken from a friend)

Total money out-flow was 550

500 for stationery

30 for transport and

20 for the lunch.

Thus, there was reduction of 50 in her wallet at the end of the day, i.e.

$$\text{change in cash in hand} = -50.$$

48.12 Money Balance Identity

At the beginning of the day: 100

At the end of the day: 50

Inflow: 500

Outflow: 550

Thus, money inflow (500) – money outflow (550)

= change in cash in hand (- 50)

$$\text{Money in-flow (500)} \equiv \text{Money out-flow (550)} + \text{change in cash in hand (-50)}$$

This gives the Money balance Identity:

48.13 The Identity in Application

- This identity is the basis of accounting methods.
- This holds good for all monetary transactions carried out during any length of time – a day, a week, a quarter or a year.

We will now use this for constructing a simple account and call it Monetary Transactions Account (MTA)

*48.14 Monetary Transactions Account (MTA)**48.15 Money In-flow*

Money in-flow is brought about by:

- Sale of products, i.e. goods & services.
- Earnings of factor compensations and property incomes
 - wages & salaries, interests, rent, dividends etc.
- Receipts of transfers and
- Incurrence of liability
- Disposal of financial assets.

Examples of incurrence of liability:

Taking loan,

Issuing of shares & bonds.

Examples of financial assets:

Deposits in bank,

Stock market shares,

Loans given to other units.

Currency or cash in hand is also financial asset

48.16 Money Out-flow

Money out-flow is brought about by:

- Purchase of products, i.e. goods & services.
- Payment of factor compensations and property incomes
- Payment of transfers and
- Extinction of liability
- Acquiring financial assets.

Like repayment of loan, buying back shares etc.

Like depositing money in bank, purchasing shares & bonds.

48.17 Monetary Transactions Account (MTA) for a household

Assume that, during an accounting period, the household

- Sold its agricultural produce: 240
- Bought goods & services (g&s): 200
- Received wages & salary (w&s): 20
- Paid w&s : 80
- Received remittances (transfers): 10
- Made transfer payments: 2
- Borrowed money: 100

48.18 MTA for a household

Note the money in-flows:

- Sold its produce: 240
- Received w&s : 20
- Received transfers: 10
- Borrowed money: 100

Thus, total money inflow: 370

Note the money out-flows:

- Purchased g&s : 200
- Paid w&s : 80
- Paid transfers: 2

Thus, total money outflow: 282

Where has the difference between inflow and outflow (= 88) gone?

Obviously, the household must hold it either in cash or as bank deposit or in other form of financial assets.

Thus, we can say: household's financial assets must have changed

48.19 MTA for a household

Thus, we get the MTA for the household

Monetary Transactions Account for a Household			
Expenditures (out-flow & holding of money)		Revenues (in-flow of money)	
200	Bought products	Sold products	240
80	Paid income	Received income	20
2	Paid transfers	Received transfers	10
88	Changes in financial assets (including change in cash in hand)	Change in (financial) liabilities	100
370	TOTAL	TOTAL	370

Note: change in financial assets is entered as 88.

Thus, totals are balanced.

48.20 Double-Entry Accounting

In general, every monetary transaction affects two cells of MTA.

Example 1: Purchase of a tractor (40) out of loan taken from bank.

Example 2: Wages (10) paid directly to bank account of household member.

Monetary Transactions Account for a Household			
Expenditures		Revenues	
40	Bought products	Sold products	
	Paid income	Received income	10
	Paid transfers	Received transfers	
	Changes in financial assets	Change in liabilities	40
	TOTAL	TOTAL	

Value of tractor purchased → 40

Wages saved in bank → 10

Receipt of wages

From loan to purchase tractor

48.21 Some Questions - MTA

State whether TRUE or FALSE.

1. Currency holding or cash in hand is NOT treated as financial asset.
2. For every monetary transaction, at least two entries are made in the MTA of a unit.
3. The sum of the money receipts should be equal to the sum of the Expenditures (including change in cash in hand).
4. The MTA provides for recording stock of assets.
5. If the 'Total' of the Revenues- and expenditures-side of MTA are unequal, there should be some missing entries.
6. Borrowing of money leads to money inflow.

Q 1. FALSE

Q 2. TRUE

Q 3. TRUE

Q 4. FALSE

Q 5. TRUE

Q 6. TRUE

48.22 Aggregating MTAs

48.23 MTAs of different units

- Like the one for the household, we can compile monetary transactions accounts (MTAs) separately for all individual residential units of an economy.
- All monetary transactions involve two parties and
 - for each party (at least) two entries are made in its MTA.
- Thus, for every transaction four entries (quadruple-entry method) are made in the system.

48.24 Aggregating MTAs

- Monetary Transaction accounts of two or more units can be aggregated by simply adding the entries of the corresponding cells.
- Aggregation over all units of an institutional sector gives the MTA of the institutional sector.
- Aggregation over all units of an economy gives the MTA of the economy.

48.25 Aggregated MTA - for resident-to-resident transactions

- Value of products sold $\equiv$ Value of products purchased
- Primary income receivable $\equiv$ Primary income payable
- Transfers receivable $\equiv$ Transfers payable
- Change in financial liabilities $\equiv$ Change in financial assets

48.26 Aggregated MTA - with external transactions

Aggregated Monetary Transactions Account for all Resident Units

Expenditure		Revenue		balance	
1235	Bought products	Sold products	1259	24	
404	Paid income	Received income	400	-4	
5	Paid transfers	Received transfers	4	-1	
1644	Sub-total: non-financial	Sub-total: non-financial	1663	19	
203	Changes in financial assets	Change in liabilities	184	-19	
1847	TOTAL	TOTAL	1847	0	

When there are external transactions, expenditure- and Revenue-side equality does NOT hold.

Balances (Expenditure-side minus Revenue-side) are results of external transactions.

Expenditures-side minus Revenues -side

48.27 Lending & Borrowing - Revisited

Aggregated Monetary Transactions Account for all Resident Units

Expenditure		Revenue		balance	
1235	Bought products	Sold products	1259	24	
404	Paid income	Received income	400	-4	
5	Paid transfers	Received transfers	4	-1	
1644	Sub-total: non-financial	Sub-total: non-financial	1663	19	
203	Changes in financial assets	Change in liabilities	184	-19	
1847	TOTAL	TOTAL	1847	0	

The balance in sub-total row shows: non-financial transactions have caused more (by 19) inflow of money than outflow and

has thus caused a larger change (by the same amount) in financial assets than that in liabilities, i.e. net lending of – 19 or net borrowing of 19.

48.28 Monetary Transaction Identities

- Value of products sold + imports

$$\equiv \text{Value of products purchased} + \text{exports}$$

- Primary income receivable from domestic units & RoW

$$\equiv \text{Primary income payable to domestic units \& RoW}$$

- Transfers receivable from domestic units & RoW

$$\equiv \text{Transfers payable to domestic units \& RoW}$$

- Change in financial liabilities + sub-total: non-financial, Revenue-side

$$\equiv \text{Change in financial assets} + \text{sub-total: non-financial, expenditure-side}$$

48.29 Some Questions - Transactions

State whether TRUE or FALSE.

1. All monetary transactions involve two parties.
2. For every transaction four entries are made in the system.
3. In an open economy, values of products bought and sold are equal.
4. The balance of non-financial transactions $\equiv$ net lending
5. Transactions identities hold good for an institutional sector as well as total economy.
6. Money borrowed and lent amongst the residents cancel out each other.

Q 1. TRUE

Q 2. TRUE

Q 3. FALSE

Q 4. TRUE

Q 5. TRUE

Q 6. TRUE

49 Integrated Transaction Accounts

49.1 Contents

- Monetary Transactions Account to basic SNA accounts
- Extension of MTA to Integrated Transaction Accounts
- Rules of Accounting

49.2 Monetary Transactions Account to Basic SNA Accounts

49.3 Aggregated Money transaction account MTA - with external transactions

Aggregated Monetary Transactions Account for all Resident Units

Expenditure		Revenue		balance
1235	Bought products	Sold products	1259	24
404	Paid income	Received income	400	- 4
5	Paid transfers	Received transfers	4	- 1
1644	Sub-total: non-financial	Sub-total: non-financial	1663	19
203	Change in fin. assets	Change in liabilities	184	- 19
1847	TOTAL	TOTAL	1847	0

Recall that

Balances (Expenditure-side minus Revenue-side) are results of external transactions.

49.4 SNA Transaction accounts and MTA

Unlike the SNA,

- the MTA does NOT have the provision of recording detailed estimates of production, consumption and the accumulation of assets.
- It only provides the value of products bought and sold without distinguishing between the transactions by purpose.
- Most importantly, the scope of MTA is limited to only monetary transactions, thus it also excludes within-unit transactions and other non-monetary transactions.

49.5 Scope of extending MTA

But, the MTA can be extended to obtain the full sequence of integrated transaction accounts, for

- the institutional sectors,
- the RoW and
- the total economy.

49.6 Extension of MTA

Extension of the MTA is required in three respects:

- first by disaggregating the MTA by institutional sectors and
- second by including all the non-monetary transactions
- third by disaggregating the broad transaction categories into further sub-categories.

The second and third kinds of extension makes the provisions for presenting the key aggregates relating to production, income, consumption and investment.

49.7 Disaggregating by Institutional Sector

The MTA for an institutional sector can be obtained by aggregating the MTAs of its individual units.

All the transaction identities do NOT hold good for each individual institutional sector, if those with the RoW are not accounted for.

However, only the last of the four identities, i.e.

Change in liabilities + sub-total: non-financial: revenue-side $\equiv$

Change in financial assets + sub-total: non-financial: expenditure-side

can be verified for the individual institutional sectors and the RoW.

49.8 Disaggregated MTAs by Institutional Sector

Sectoral Monetary Transactions Account and Transactions with RoW

Sectoral Monetary Transactions Account and Transactions with RoW								
Transaction category	NFC	FC	GG	HH	NPISHs	Dom. Econ.	RoW	total
revenue								
Sold products	847	155	7	248	2	1259	256	1515
Received income	131	75	26	157	11	400	65	465
Received transfers	0	0	3	0	1	4	2	6
Sub-total: non-financial	978	230	36	405	14	1663	323	1986
→ Change in liabilities	92	57	27	4	4	184	51	235
Total (= sub-tot. + item 4)	1070	287	63	409	18	1847	374	2221
expenditure								
Bought products	707	112	23	392	1	1235	280	1515
Paid income	215	151	29	7	2	404	61	465
Paid transfers	2	1	0	2	0	5	1	6
Sub-total: non-financial	924	264	52	401	3	1644	342	1986
→ Change in fin. assets	146	23	11	8	15	203	32	235
Total (= sub-tot. + item 4)	1070	287	63	409	18	1847	374	2221

Note that: For all the sectors,

sub-total: non-financial, revenue-side + change in liabilities =

sub-total: non-financial, expenditure-side + change in financial asset

49.9 RoW Account

The rest of the world (RoW) account covers transactions between resident and non-resident institutional units.

This is presented from the viewpoint of the RoW.

49.10 RoW Account Entries

Sectoral Monetary Transactions Account and Transactions with RoW

(Refer to the table above)

Note that

‘sold products’ of RoW is imports to domestic economy

‘bought products’ of RoW is exports to domestic economy

A revenue head for the RoW is an expenditure for the total economy and vice versa. Thus, ‘paid income’ and ‘paid transfers’ of the RoW are respectively ‘received income’ and ‘received transfers’ of the domestic economy.

49.11 Some Questions – MTA to SNA

State whether TRUE or FALSE.

1. MTAs have the provision of recording detailed estimates of production, consumption and the accumulation of assets.
2. The MTAs can be extended to obtain the full sequence of transaction accounts.
3. The scope of MTAs is limited to only monetary transactions.
4. All the transaction identities do NOT hold for institutional sector accounts.
5. The RoW account is presented from the view point of the non-residents.

6. Payment of transfers to non-residents is shown as ‘transfer paid’ in the RoW account.
7. Difference between changes in financial assets and liabilities of RoW is the economy’s net lending / borrowing.
- Q 1. FALSE
- Q 2. TRUE
- Q 3. TRUE
- Q 4. FALSE
- Q 5. TRUE
- Q 6. FALSE
- Q 7. TRUE

49.12 Extension of MTA to Elements of Transaction Accounts

49.13 Extension of MTA

The economy-level MTA is now extended:

- first by including the non-monetary transactions,
- second by breaking down use of products into its components
- third by breaking down (primary) income into its components and
- fourth by breaking down transfers into ‘current’ and ‘capital’ components.

49.14 Inclusion of Non-monetary transactions – products, revenue-side

The SNA production boundary includes the following non-monetary transactions:

- production for own final use (23), and
- non-market production (43) by general government, CB and NPISHs.

Thus, on the revenue-side of the MTA, to ‘products sold’, these are added.

Together, these now represent gross value of output (1325)

revenue	
gross value of output (GVO)	1325
Products sold	1259
For own use	23
Non-market output	43

49.15 Disaggregating by purpose – products, expenditure-side

Expenditure (Use) of products is given by

purchase of products (1235) + value of products for own use (23)

+ value of non-market output (43) = 1301

These are used for Final consumption, Intermediate consumption and Capital formation

Thus, on the expenditure-side, use of products is disaggregated into these 3 categories.

expenditure	
Use of products	1301
Intermediate consumption	870
Final consumption	408
Capital formation	23

49.16 Disaggregating Income into Components

The second category of transactions – payment and receipt of income – in the MTA is then disaggregated by its components

- Remuneration of employees
- Property income
- Production (t-s) & import duties

These include both received from and paid to the RoW.

expenditure		revenue	
404	Paid income	Received income	400
153	Remuneration of employees	Remuneration of employees	152
121	Property income	Property income	118
130	Production (t-s) & import duties	Production (t-s) & import duties	130

49.17 Distinguishing Current & Capital Transfers

The third category of transactions – payment and receipt of transfers – in the MTA is then disaggregated into

- Current transfers and
- Capital transfers

These are both received from and paid to the RoW.

expenditure		revenue	
5	Transfers paid	Transfers received	4
3	Current transfers	Current transfers	3
2	Capital transfers	Capital transfers	1

49.18 Re-modified MTA

Re-modified Monetary Transactions Account

Re-modified Monetary Transactions Account			
expenditure		revenue	
Total economy	Transaction category	Transaction category	Total economy
1301	1. Use of products	1. GVD _{mp}	1325
870	Intermediate consumption	Products sold (at basic price)	1259
408	Final consumption	For own use	23
23	Capital formation	Non-market	43
		(t-s) on products & import duties	126
404	2. Paid income	2. Received income	400
153	Remuneration of employees	Remuneration of employees	152
121	Property income	Property income	118
130	Production (t-s) & import duties	Production (t-s) & import duties	130
5	3. Paid transfers	3. Received transfers	4
3	Current transfers paid	Current transfers received	3
2	Capital transfers paid	Capital transfers received	1
1710	Sub-total: non-financial	Sub-total: non-financial	1729
203	4. Change in financial assets	4. Change in liabilities	184
1913	Total (= sub-total + item 4)	Total (= sub-total + item 4)	1913

Re-modified MTA

Note that, as compared to Table 3.1 (in an earlier slide):

- (a) Here the entry in revenue-side, 'products sold' is unchanged, only the entries 'for own use' (23) and 'non-market' (43) are in addition;
- (b) Thus, the expenditure-side entry (1301) is higher by 66 (= 23 + 43) than the corresponding entry (1235)
- (c) Income paid (404) and received (400) have not changed, only disaggregated into components;
- (d) Similarly, transfer paid (5) and received (4) have not changed, only divided into components;
- (e) Most importantly, the difference (1729 – 1710 = 19) between the revenue- and expenditure-side sub-totals does NOT change.

49.19 MTA to Main Structure of SNA Transaction Accounts

49.20 MTA to Transaction Accounts

We can now obtain the main structure of the SNA transaction sequence accounts from the regrouped entries of the revenue- and expenditure-side of the MTA in the earlier slide.

Let's assume D&D = 10, for presenting the aggregate in net terms.

We will construct the accounts one-by-one in the next few slides.

49.21 Production Account from MTA

Re-modified Monetary Transactions Account and Transactions with RoW

SNA Sequence of Transaction Accounts in net terms

Production Account	
GVO	1325
bp	870 IC
Products sold (at basic price)	1259
For own use	23
Non-market	43
(t-s) on products & import duties	126
GDP	455
D&D	10
NDP	445

MTA to Transaction Accounts				MTA to Transaction Accounts			
Production Account from MTA				Generation of Income Account from MTA			
Re-modified Monetary Transactions Account and Transactions with RoW				Table 3.4: Re-modified Monetary Transactions Account and Transactions with RoW			
expenditure	Transaction category	revenue	Total economy	expenditure	Transaction category	revenue	Total economy
1301	1. Use of products	1. GVO _{bp}	1325	1301	1. Use of products	1. GVO _{bp}	1325
870	Intermediate consumption	Products sold (at basic price)	1259	870	Intermediate consumption	Products sold (at basic price)	1259
408	Final consumption	For own use	23	408	Final consumption	For own use	23
23	Capital formation	Non-market	43	23	Capital formation	Non-market	43
		(t-s) on products & import duties	126			(t-s) on products & import duties	126
404	2. Paid income	2. Received income	400	404	2. Paid income	2. Received income	400
153	Remuneration of employees	Remuneration of employees	152	153	Remuneration of employees	Remuneration of employees	152
121	Property income	Property income	118	121	Property income	Property income	118
130	Production (t-s) & import duties	Production (t-s) & import duties	130	130	Production (t-s) & import duties	Production (t-s) & import duties	130
5	3. Paid transfers	3. Received transfers	4	5	3. Paid transfers	3. Received transfers	4
3	Current transfers paid	Current transfers received	3	3	Current transfers paid	Current transfers received	3
2	Capital transfers paid	Capital transfers received	1	2	Capital transfers paid	Capital transfers received	1
1710	Sub-total: non-financial	Sub-total: non-financial	1729	1710	Sub-total: non-financial	Sub-total: non-financial	1729
203	4. Change in financial assets	4. Change in liabilities	184	203	4. Change in financial assets	4. Change in liabilities	184
1913	Total (= sub-total + item 4)	Total (= sub-total + item 4)	1913	1913	Total (= sub-total + item 4)	Total (= sub-total + item 4)	1913

49.22 Generation of Income Account from MTA

Table 3.4: Re-modified Monetary Transactions Account and Transactions with RoW

Generation of Earned Income Account

$$\text{NDP} = 445$$

$$\text{Remuneration of employees} = 153$$

$$\text{Production (t-s) \& import duties} = 130$$

$$\text{OS} + \text{MI (net)}$$

$$= 445 - 153 - 130$$

49.23 MTA to Allocation of Primary Income Account

Table 3.4: Re-modified Monetary Transactions Account and Transactions with RoW

Allocation of Earned Income Account

$$OS + MI \text{ (net)} = 162$$

$$\text{Remuneration of employees} = 152$$

$$\text{Property income} = 118$$

$$\text{Production (t-s) \& import duties} = 130$$

$$\text{Property income} = 121$$

$$NNI$$

$$= 162 + 152 + 118 + 130 - 121$$

$$= 441$$

MTA to Transaction Accounts			
MTA to Allocation of Primary Income Account			
Table 3.4: Re-modified Monetary Transactions Account and Transactions with RoW			
expenditure	Transaction category	revenue	Transaction category
Total economy	Transaction category	Transaction category	Total economy
1301	1. Use of products	1. GVO _{exp}	1325
870	Intermediate consumption	Products sold (at basic price)	1259
408	Final consumption	For own use	23
23	Capital formation	Non-market	43
		(t-s) on products & import duties	126
404	2. Paid income	2. Received income	400
153	Remuneration of employees	Remuneration of employees	152
121	Property income	Property income	118
130	Production (t-s) & import duties	Production (t-s) & import duties	130
5	3. Paid transfers	3. Received transfers	4
3	Current transfers paid	Current transfers received	3
2	Capital transfers paid	Capital transfers received	1
1710	Sub-total: non-financial	Sub-total: non-financial	1729
203	4. Change in financial assets	4. Change in liabilities	184
1913	Total (= sub-total + item 4)	Total (= sub-total + item 4)	1913

From generation of earned income account
 $= 162 + 152 + 118 + 130 - 121$
 Allocation of Earned Income Account
 $OS + MI \text{ (net)}$
 Remuneration of employees
 Property income
 Production (t-s) & import duties
 NNI

MTA to Transaction Accounts			
MTA to Secondary Distribution of Income Account			
Table 3.4: Re-modified Monetary Transactions Account and Transactions with RoW			
expenditure	Transaction category	revenue	Transaction category
Total economy	Transaction category	Transaction category	Total economy
1301	1. Use of products	1. GVO _{exp}	1325
870	Intermediate consumption	Products sold (at basic price)	1259
408	Final consumption	For own use	23
23	Capital formation	Non-market	43
		(t-s) on products & import duties	126
404	2. Paid income	2. Received income	400
153	Remuneration of employees	Remuneration of employees	152
121	Property income	Property income	118
130	Production (t-s) & import duties	Production (t-s) & import duties	130
5	3. Paid transfers	3. Received transfers	4
3	Current transfers paid	Current transfers received	3
2	Capital transfers paid	Capital transfers received	1
1710	Sub-total: non-financial	Sub-total: non-financial	1729
203	4. Change in financial assets	4. Change in liabilities	184
1913	Total (= sub-total + item 4)	Total (= sub-total + item 4)	1913

From allocation of earned income account
 Income transfer (other than STK) Account
 NNI
 Current transfers payable
 Current transfers receivable
 GNDI

49.24 MTA to Secondary Distribution of Income Account

Table 3.4: Re-modified Monetary Transactions Account and Transactions with RoW

Income transfer (other than STK) Account

$$NNI = 441$$

Current transfers receivable

Current transfers payable

$$GNDI = 441$$

49.25 MTA to Use of Disposable Income Account

Table 3.4: Re-modified Monetary Transactions Account and Transactions with RoW

MTA to Transaction Accounts

MTA to Capital Account

Table 3.4: Re-modified Monetary Transactions Account and Transactions with RoW

expenditure		revenue	
Total economy	Transaction category	Transaction category	Total economy
1301	1. Use of products	1. GVO _{mp}	1325
870	Intermediate consumption	Products sold (at basic price)	1259
408	Final consumption	For own use	23
23	Capital formation	Non-market	43
		(t-s) on products & import duties	126
404	2. Paid income	2. Received income	400
153	Remuneratio of employees	Remuneratio of employees	152
121	Property income	Property income	118
130	Production (t-s) & import duties	Production (t-s) & import duties	130
5	3. Paid transfers	3. Received transfers	4
3	Current transfers paid	Current transfers received	3
2	Capital transfers paid	Capital transfers received	1
1710	Sub-total: non-financial	Sub-total: non-financial	1729
203	4. Change in financial assets	4. Change in liabilities	184
1913	Total (= sub-total + item 4)	Total (= sub-total + item 4)	1913

Capital Account

changes in assets		changes in liability & net worth	
		Net savings	?
?	Capital transfers payable	Capital transfers receivable	?
?	Gross Capital formation		
- 10	D&D		
?	Net lending / borrowing		

From use of disposable income account

Note that net lending / borrowing is 19 – same as the difference between changes in financial asset and liabilities.

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Use of Disposable Income Account

$$\text{NNDI} = 441$$

$$\text{Final consumption} = 408$$

$$\text{Net savings} = 33$$

49.26 MTA to Capital Account

Table 3.4: Re-modified Monetary Transactions Account and Transactions with RoW
Capital Account

changes in assets	changes in liability & net worth
Net savings	33
Capital transfers receivable	1
Capital transfers payable	2
Gross Capital formation	23
D&D	-10
Net lending / borrowing	19

49.27 Some Questions - MTA

State whether TRUE or FALSE.

1. The difference between sums of revenue- and expenditure-side of non-financial transactions is always equal to net borrowing / lending.

2. Net borrowing / lending is always equal to change in financial assets minus change in financial liabilities.
 3. The balancing item net borrowing / lending is affected by non-monetary transactions.
 4. The crop output 'for own use' is recorded in use of income account.
 5. Depreciation reported in company accounts is taken as the CFC.
 6. Import duties are recorded in the production account of an economy.
 7. Goods and services produced in an economy must be (finally) consumed, used for capital formation or exported.
- Q 1. TRUE
- Q 2. TRUE
- Q 3. FALSE
- Q 4. TRUE
- Q 5. FALSE
- Q 6. TRUE
- Q 7. TRUE

49.28 Rules of Accounting

49.29 MTAs of different units (1)

The quadruple-entry accounting, in principle, ensures matched reporting by the two involved parties. For a transaction (say sale of products) between parties 'A' and 'B', it is necessary to record

- the same value for both of them and
- it for the same accounting period.

Otherwise, this will lead to inconsistency.

49.30 MTAs of different units (2)

Two important rules necessary for consistency of accounts are:

- Flows and stocks must be recorded consistently with respect to their valuation.
- Entries must be at current value on the market or at its closest equivalent.
- Flows and stocks must be recorded consistently with respect to timing.

49.31 Valuation of transactions

The rules of valuation:

- All monetary exchange transactions, are recorded at the prices actually paid by the buyer.
- For non-monetary transactions, valuation according to market-price equivalents is taken as an approximation to market prices.
- If there is no appropriate market for the non-monetary exchange transactions of goods or services, the valuation is done as sum of costs.
- Exports and imports are valued free on board (fob).

49.32 Time of Recording

Rules for time of recording:

- For transaction of products: The moment of change in economic ownership.
- For financial assets & liabilities: The moment of change in economic ownership.
- Receipt & payment of primary income and transfers: the period they become due.
- Output and intermediate consumption: the period when the production takes place.

49.33 Some Questions - Transactions

State whether TRUE or FALSE.

1. All non-monetary transactions are valued at cost.
2. Transactions of primary income and transfers are recorded in the accrual basis.
3. Monetary exchange transactions should be recorded at the average prices prevailing during the accounting year.
4. All flows and stocks are recorded at current value
5. All transactions identities hold good for an institutional sector as well as total economy.
6. Output and intermediate consumption should always be recorded respectively at the selling and buying prices.

Q 1. FALSE

Q 2. TRUE

Q 3. FALSE

Q 4. TRUE

Q 5. TRUE

Q 6. FALSE